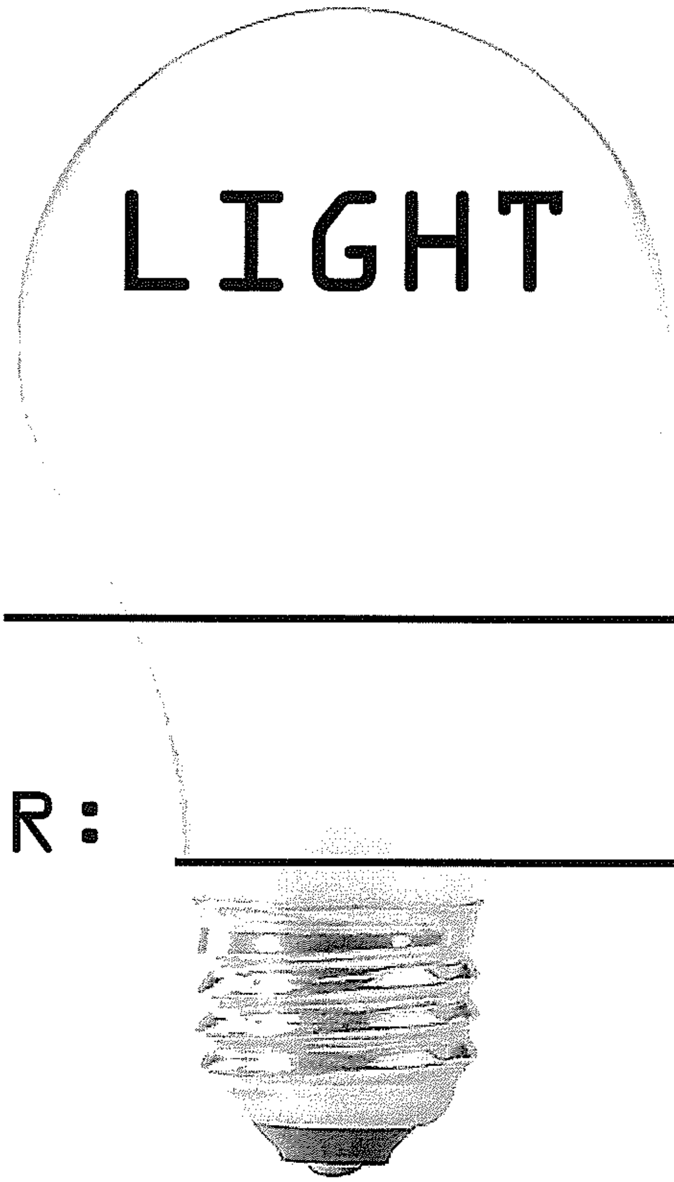
YEAR 10 SCIENCE

LIGHT



NAME: ______________________________

TEACHER: __________________________

HISTORY OF LIGHT

➔ Over the centuries, our view of light has changed dramatically. The first real theories about light came from the ancient Greeks. Many of these theories sought to describe light as a **ray** -- a straight line moving from one point to another. Pythagoras, best known for the theorem of the right-angled triangle, proposed that vision resulted from light rays emerging from a person's eye and striking an object. Epicurus argued the opposite: Objects produce light rays, which then travel to the eye. Other Greek philosophers -- most notably Euclid and Ptolemy -- used ray diagrams quite successfully to show how light bounces off a smooth surface or bends as it passes from one transparent medium to another.

Pythagoras
D.O.B 580-572 BC
D.O.D 500-490 BC
Birthplace - France

Laser eyes

➔ Arab scholars took these ideas and honed them even further, developing what is now known as **geometrical optics** -- applying geometrical methods to the optics of lenses, mirrors and prisms. The most famous practitioner of geometrical optics was **Ibn al-Haytham**, who lived in present-day Iraq between A.D. 965 and 1039. Ibn al-Haytham identified the optical components of the human eye and correctly described vision as a process involving light rays bouncing from an object to a person's eye. The Arab scientist also invented the pinhole camera, discovered the laws of refraction and studied a number of light-based phenomena, such as rainbows and eclipses.

Ibn al-Haytham
D.O.B 965 CE
D.O.D 1040 CE
Place of birth - Iraq

Christiaan Huygens
D.O.B 1629 CE
D.O.D 1695 CE
Place of birth - France

➔ By the 17th century, some prominent European scientists began to think differently about light. One key figure was the Dutch mathematician-astronomer **Christiaan Huygens**. In 1690, Huygens published his "Treatise on Light," in which he described the **undulatory theory**. In this theory, he speculated on the existence of some invisible medium which filled empty space between objects. He further speculated that light forms when a luminous body causes a series of waves or vibrations in this ether. Those waves then advance forward until they encounter an object. If that object is an eye, the waves stimulate vision.

→ This stood as one of the earliest, and most eloquent, wave theories of light. Not everyone embraced it. Isaac Newton was one of those people. In 1704, Newton proposed a different take -- one describing light as particles. After all, light travels in straight lines and bounces off a mirror much like a ball bouncing off a wall. No one had actually seen particles of light, but even now, it's easy to explain why that might be. The particles could be too small, or moving too fast, to be seen, or perhaps our eyes see right through them.

Isaac Newton
D.O.B 1643
D.O.D 1727
Place of birth - England

→ As it turns out, all of these theories are both right and wrong at once. And they're all useful in describing certain behaviors of light.

→ Imagining light as a ray makes it easy to describe.

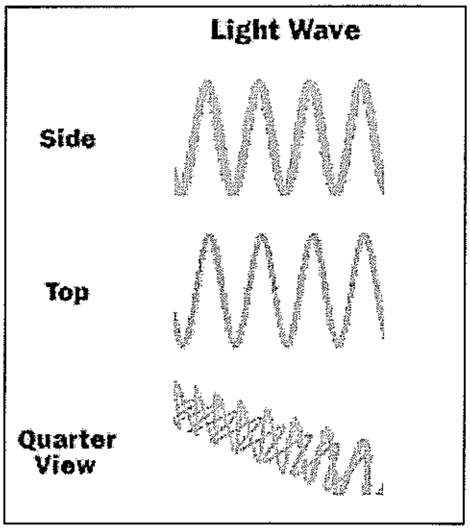
Reflection – a light ray strikes a smooth surface, such as a mirror, and bounces off. A reflected ray always comes off the surface of a material at an angle equal to the angle at which the incoming ray hit the surface (law of reflection).

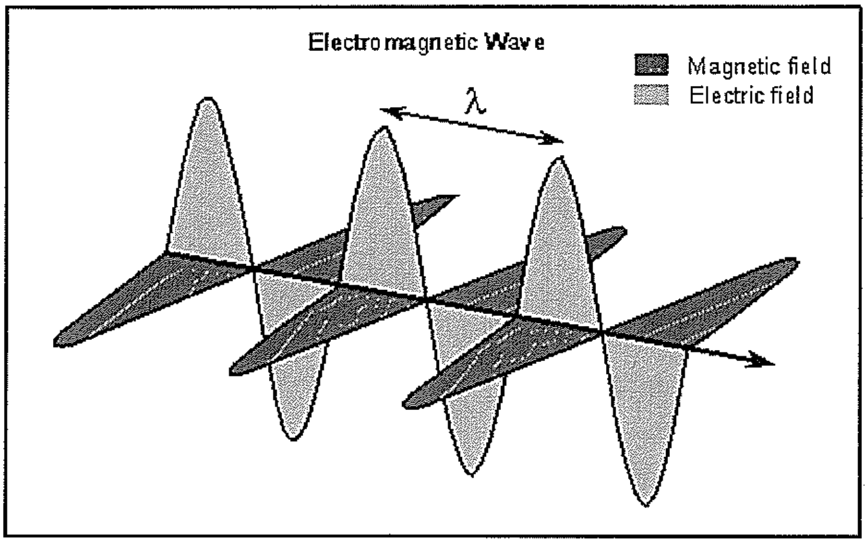
Scattering – we live in an imperfect world and not all surfaces are smooth, when light strikes a rough surface, incoming light rays reflect at all sorts of angles because the surface is uneven. This scattering occurs in many of the objects we encounter every day. This piece of paper is a good example. You can see just how rough it is if you look at it under a microscope. When light hits paper, the rays are reflected in all directions. Therefore, you can read these words regardless of the angle at which your eyes view the surface.

Refraction – this occurs when a ray of light passes from one transparent medium (for example, air) to a second transparent medium (for example, water). When this happens, light changes speed and the light ray bends.

→ Light does travel in straight lines, but these straight lines are made up of a **wave formation**. This was discovered by Thomas Young, an English physician and physicist, in 1801. Incidentally, the bending of a light wave accounts for some of the visual phenomena we often encounter, such as mirages. A **mirage** is an optical illusion caused when light waves moving from the sky toward the ground are bent by the heated air.

➔ In the 1860's, Scottish physicist James Clerk Maxwell put the cherry on top of the light-wave model when he formulated the theory of **electromagnetism**. Maxwell described light as a very special kind of wave – one composed **of electric and magnetic fields**. The fields vibrate at right angles to the direction of the movement of the wave, and at right angles to each other. Because light has both electric and magnetic fields, it is also referred to as **electromagnetic radiation**.



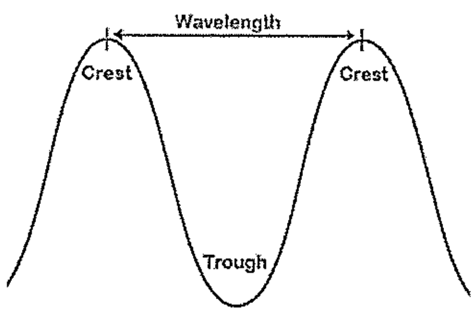


Medium: a substance that makes the transfer of energy from one location to another possible, especially through waves.

➔ Electromagnetic radiation **doesn't need a medium** to travel through and when it travels through a vacuum, it moves at **300, 000 kilometers per second**. Scientists refer to this as the **speed of light**, one of the most important numbers in physics.

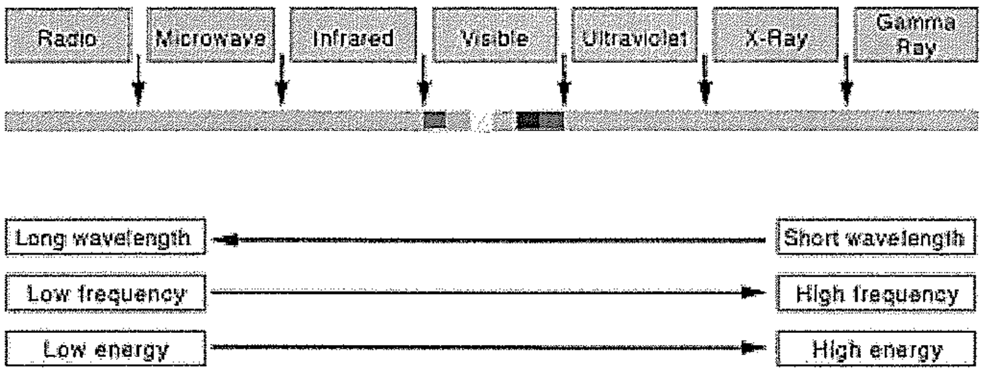
➔ Once Maxwell introduced the concept of electromagnetic waves, everything clicked into place. Scientists could now develop a complete working model of light using terms and concepts, such as **wavelength** and **frequency**, based on the structure and function of waves.

➔ The size of a wave is measured as its **wavelength**, which is the distance between peak to peak or trough to trough. The wavelengths of the light we can see range from 400 to 700 nanometers (billionths of a meter). But the full range of wavelengths included in the definition of electromagnetic radiation extends from 0.1 nanometers (as in gamma rays), to centimeters and meters (as in radio waves).



→ Light waves also come in many **frequencies**. The frequency is the number of waves that pass a point in space during any time interval, usually one second. We measure it in units of cycles (waves) per second, or **hertz**. The frequency of visible light is referred to as colour, and ranges from 430 trillion hertz, seen as red, to 750 trillion hertz, seen as violet. Again, the full range of frequencies extends beyond the visible portion, from less than 3 billion hertz (as in radio waves), to greater than 3 billion hertz (3×10^{19}) as in gamma rays.

→ Of visible light, violet has the most energy and red the least. The whole range of frequencies and energies is known as the **electromagnetic spectrum**. Remember that the diagram below isn't drawn to scale and that visible light only occupies one-thousandth of a percent of the spectrum.



→ Maxwell's theory of electromagnetic radiation was so elegant and predictive that many physicists in the 1890's thought that there was nothing more to say about light and how it worked. Then, on December 14, 1900, Max Planck came along and introduced a stunningly simple, yet strangely unsettling concept, that light must carry **energy** in some quantities.

→ Those quantities, he proposed, must be units of the basic energy increment, hf, where h is the universal constant now known as Planck's constant and **f** is the **frequency** of the radiation.

→ Albert Einstein added to Planck's theory in 1905 when he studied the photoelectric effect. He shone ultraviolet light on the surface of a metal, he was able to detect electrons being emitted from the surface. Einstein explained that light existed as tiny particles called **photons** which could bounce from off metal surfaces.

→ Today physicists accept the dual nature of light. Light not only moves in waves, but is also made up of **particles**.

__Light__: electromagnetic radiation to which the organs of sight react. Also called luminous energy or radiant energy.

Fill in the missing blanks.

➔ Light is a form of e______________________ e_____________ that can be detected by our eyes.

➔ This invisible light can be controlled by mirrors and lenses and we can see its effects. The rainbow, shadows, moving images on a movie screen are some of the effects of l________ we can see.

➔ Light travels in the form of ___________ through air, water, solids and even through a _________________ (like in space).

➔ Light travels at ______________________ kilometers per second. It can ___________ (circle around) the Earth 8 times in one second.

➔ Light travels in _________________ lines, but if it hits another medium it ________________.

➔ The symbol 'f' represents ______________________________.

➔ The type of waves with the lowest frequency are __________________________.

➔ The type of waves with the highest frequency are __________________________.

➔ The type of waves with the lowest energy are ________________________________.

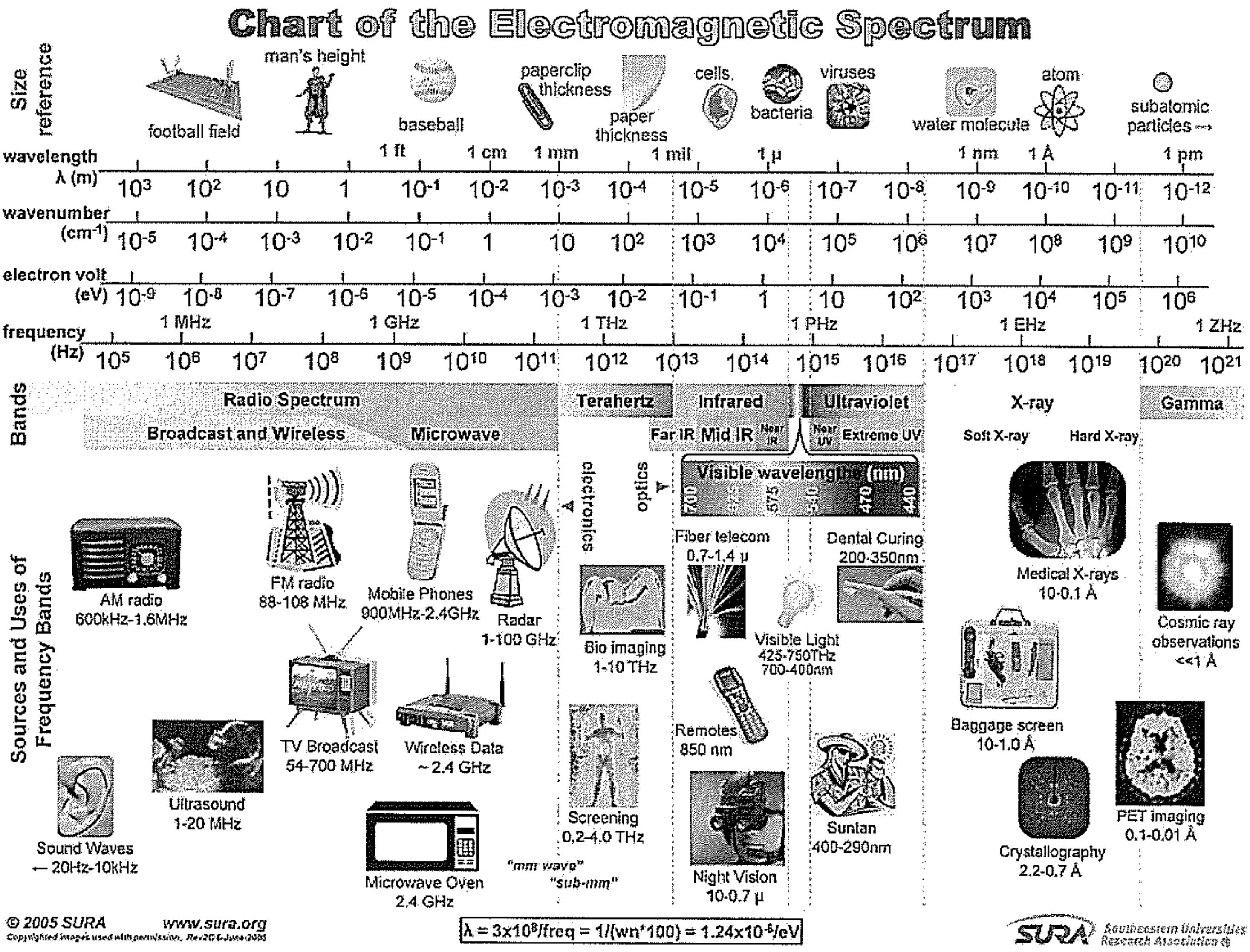
➔ The type of waves with the highest energy are ____________________________.

➔ The type of waves with the longest wavelengths are __________________________.

➔ The type of waves with the shortest wavelengths are ___________________________.

THE ELECTROMAGNETIC SPECTRUM

Wave type	How it is detected/used
Radio waves	
Microwaves	
Infra-red (heat)	
Light (visible)	
Ultra-violet	
X-rays	
Gamma rays	



TYPES OF OBJECTS

➔ **Luminous** objects **produce their own light**, e.g. the Sun, stars, a lighted candle, fire, light bulb.

- **Bioluminescence** animals **produce their own light**, e.g. glow worms, fire flies.

➔ **Non-luminous** objects do not produce their own light but can reflect light, e.g. mirrors and the moon.

➔ **Opaque** materials **do not allow light** to pass through them, e.g. wood and paper but a shiny mirror can reflect light rays (bounce back). Either **blocks or absorbs** all the light.

➔ **Transparent** material **transmits light** (allows all light to pass through it), e.g. air, water, glass, clear plastic.

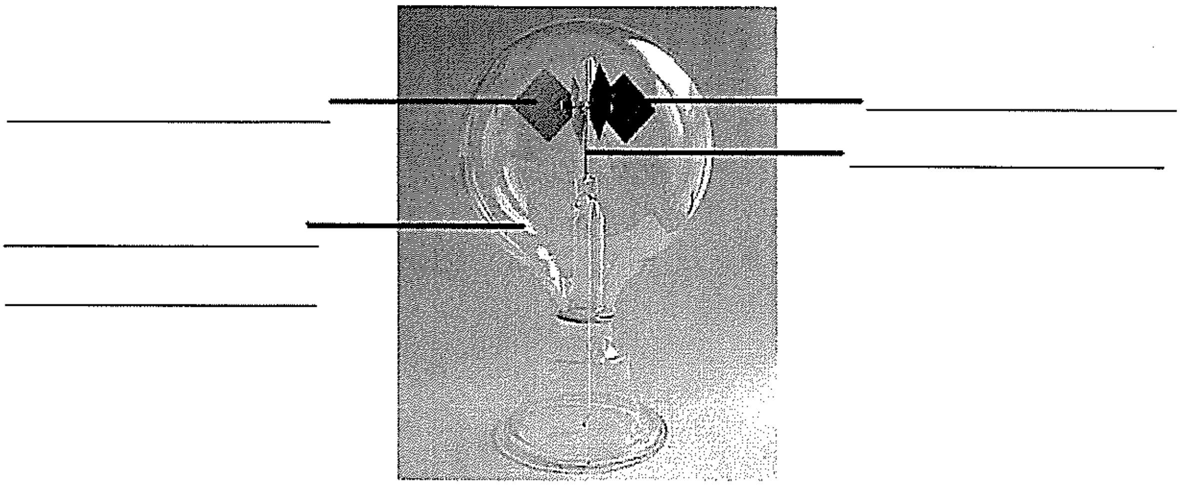
➔ **Translucent** objects let **some light** pass through it, e.g. tissue paper, frosted glass, baking paper.

Demonstration – shine a torch through a piece of glass, baking paper and a wooden board.

Demonstration – Radiometer

Materials – radiometer, torch

This instrument was designed to detect and measure radiant energy. A simple type, devised by Sir William Crookes, consists of four vanes of aluminium foil that revolve on a needle point within a glass globe from which most of the air has been removed. One face of each vane is blackened, while the other side is bright. When placed in sunlight or near any light, the vanes rotate in the direction faced by the bright sides. The black surface absorbs the heat and radiates (gives off) heat while the bright surface reflects heat. The greater heating of the blackened surfaces, according to one theory, causes nearby air



REFLECTION

➔ Practically everything we see is visible to us by reflected light. Only luminous objects like the sun, a flame or a glowing electric light filament are visible by direct light.

➔ When a ray of light strikes a smooth surface at an angle, it is reflected from the surface at an equal angle. This is known as the **Law of reflection**.

➔ When the ray of light strikes the surface at an angle of 90 degrees, it is reflected straight back into itself.

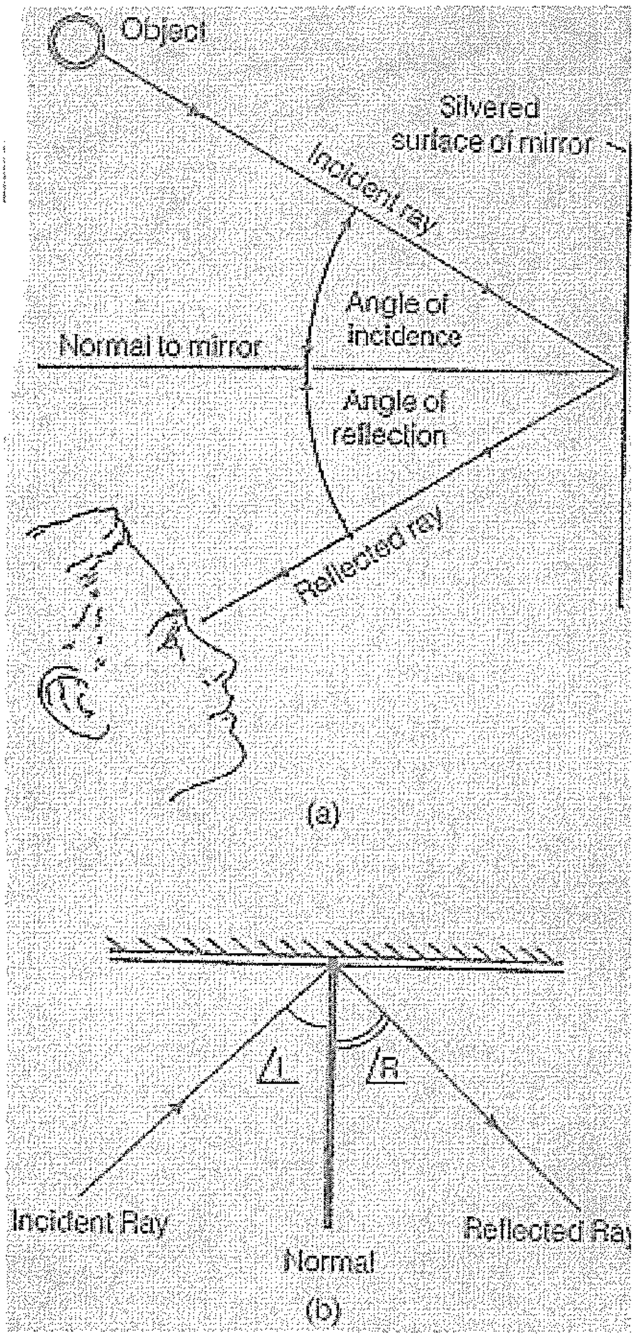
➔ **Incident ray** – is the ray that strikes the surface.

➔ **Reflected ray** – is the ray that is reflected off the surface.

➔ **Normal** – is a perpendicular line drawn at the point where the incident ray strikes the surface.

➔ **Angle of incidence** – is the angle between the incident ray and the normal.

➔ **Angle of reflection** – is the angle between the reflected ray and the normal.



Laws of reflection

- *The angle of incidence is equal to the angle of reflection.*
- *For a plane mirror the incident ray, the reflected ray and the normal are on the same plane*

Regular and irregular or diffused reflection

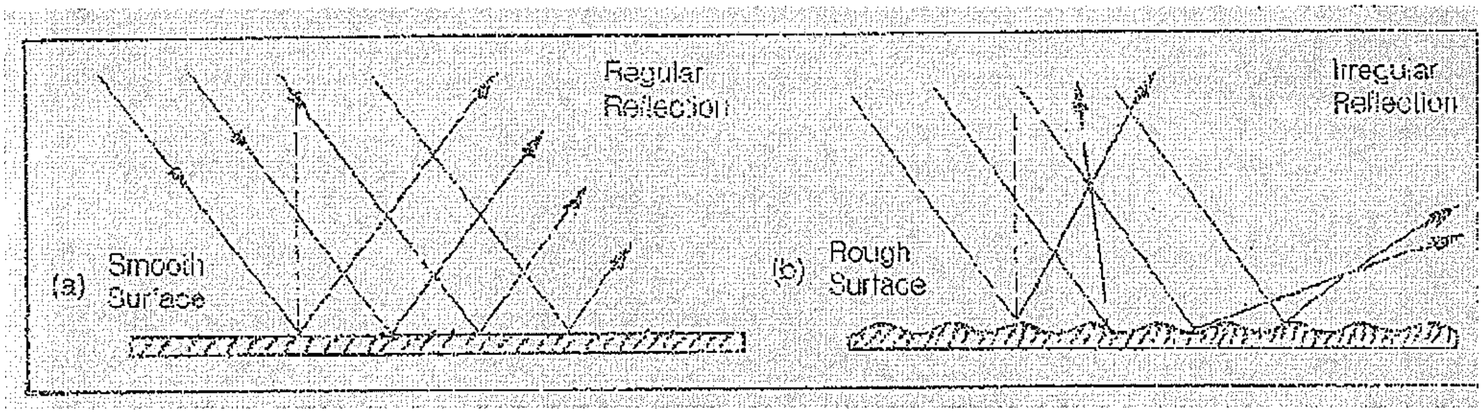
➔ A **smooth** surface reflects rays of light to the eye at about the same angle, while a **rough** surface reflects these rays at many different angles.

➔ Look at the diagram below, you can see that each ray individually obeys the lays of reflection.

➔ Light reflected from a smooth surface is said to undergo **regular reflection**.

➔ Light reflected from a rough surface is said to undergo irregular or **diffused reflection**.

List three smooth surfaces that would undergo regular reflection.



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List three rough surfaces that would undergo irregular reflection.

____

Fill in the blanks

In a plane mirror, the angle of ____________ is always ________ to the angle of reflection. This fact is known as the ________ ___ ______________.

Smooth ____________ surfaces produce regular reflection. ________ reflecting surfaces produce ______________ or ______________ reflection.

MIRRORS

Types of mirrors

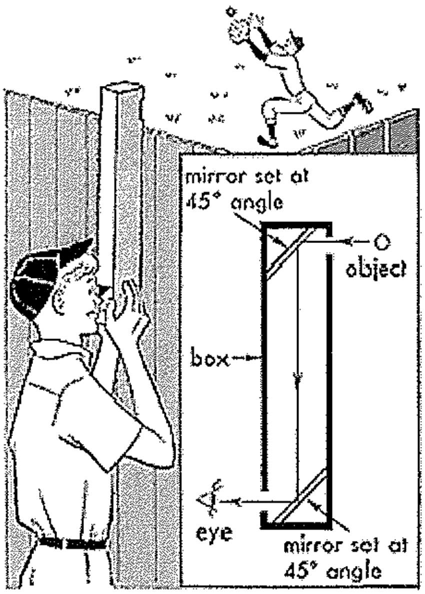
➔ Mirrors are light reflecting devices, usually made of glass with a thin coating of a reflective material on one side. There are three classes of mirrors; plane, concave and convex. Cheaper mirrors are coated behind the sheet of glass. High quality precision mirrors are coated on the front face of the glass. These do not distort light and are used in precision instruments (e.g. dentist and surgery tools).

➔ In cheaper mirrors, mercury and aluminium are usually sued. Silver and gold are used in making expensive precision mirrors.

FLAT MIRRORS

➔ **Plane mirrors** – **flat** in shape.

➔ Flat or plane mirrors are commonly used in our homes. When you see yourself in a plane mirror, your image appears to be the **same size** as yourself and it is **upright**. However, it is reversed from side to side (**laterally reversed**). Your image in the mirror also seems to be as far behind the mirror as you are in front.



➔ Remember, the image you see in the mirror is not there at all, but is actually a **reflection** detected by the retina of your eye. This image **cannot be produced on a screen**.

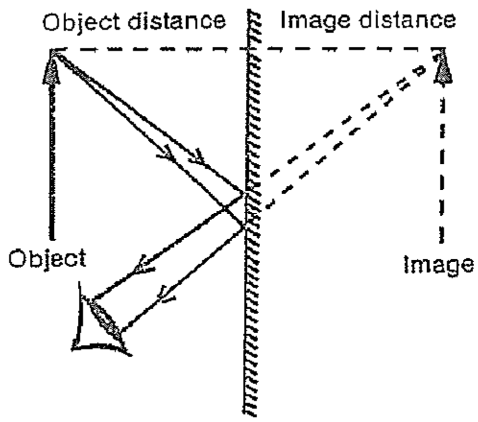
➔ Such images are called **virtual or false images**. Images which can be formed on a screen using certain types of curved mirrors and lenses are called real images.

➔ Plane mirrors are used in bathrooms and in periscopes.

Where would periscopes be used and why?

____

PLANE MIRROR



A plane mirror produces an upright, virtual image that is the same size as the object. The object and image distances are equal.

➔ Look at the image on the left. a) Shows the reversal of the image relative to the object.

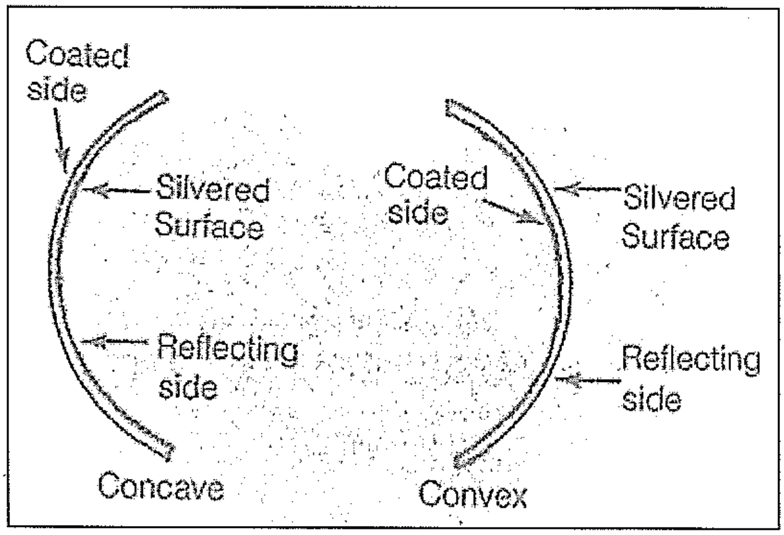
➔ b) shows that the image formed by a plane mirror is as far behind the mirror as the object is in front of it.

Why do ambulance vehicles have the word 'ambulance' printed backwards?

SPHERICAL MIRRORS

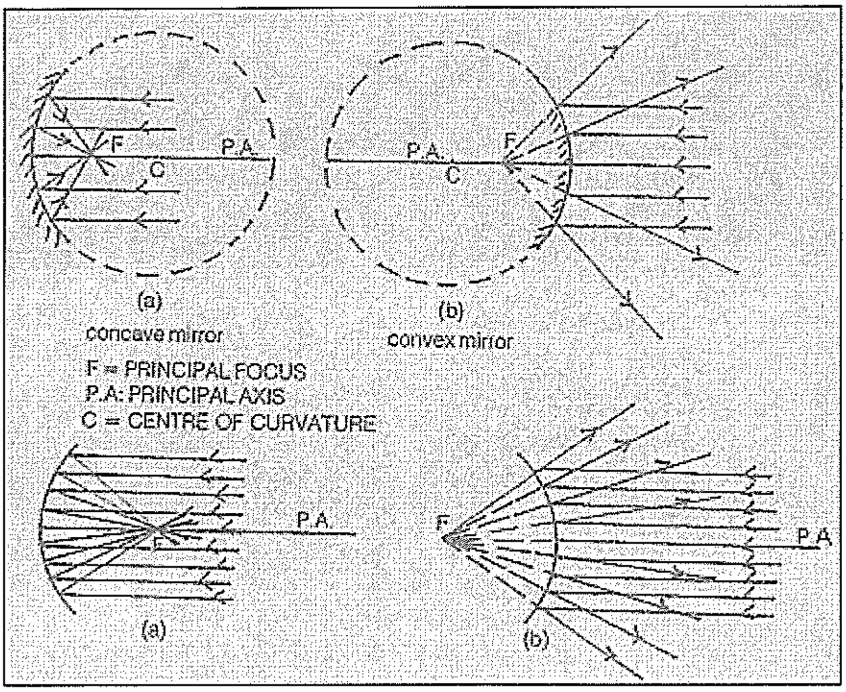
➔ Two types of spherical mirrors are the **concave** mirrors and the **convex** mirrors. Both are spherical because they have curved reflecting surfaces.

➔ Each of these mirrors is curved in such a way that each forms part of the surface of a **sphere**. The centre of this imaginary sphere is called the **centre of curvature**.



➔ A convex mirror reflects light rays outwards and a concave mirror reflects light inwards.

➔ The straight line passing through the mid point of a curved mirror and its centre of curvature is known as the principal axis.



➔ Look at the image on the left.
a) shows rays of light falling on a concave mirror converging or reflecting inwards.

b) shows rays of light falling on a convex mirror diverging or reflecting outwards.

➔ The **principal focus** is found to be midway between the centre of curvature and the centre of the mirror. The distance between the principal focus and the centre of the mirror is called the **focal length**.

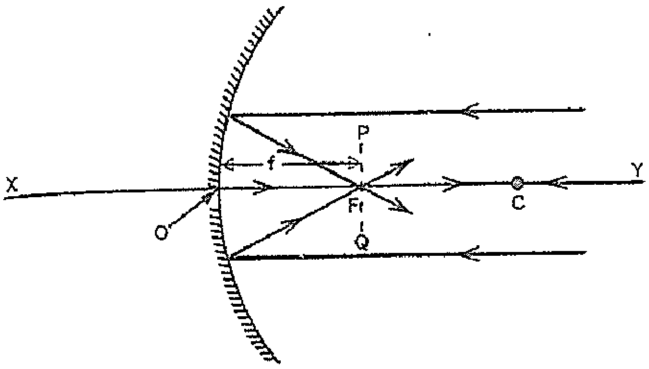
➔ The following rules must be followed when diagrams like the one above are drawn.

1) Rays drawn parallel to the principal axis, will, after reflection, pass through the principal focus.

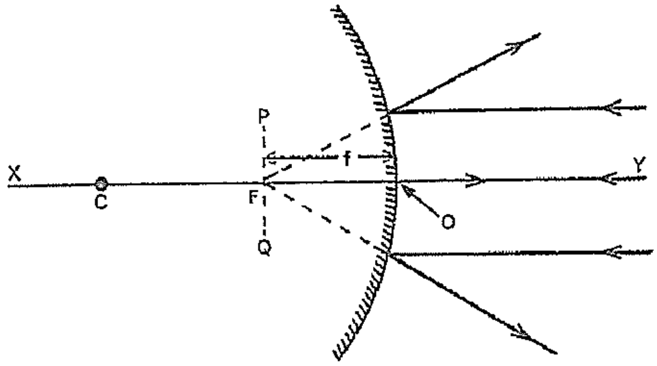
2) Rays passing through the principal focus, will, after reflection, travel parallel to the principal axis.

3) Rays passing through the centre of curvature will strike the mirror at right angles and retrace its path.

TERMS USED IN CONNECTION WITH MIRRORS



Concave mirror



Convex mirror

Term		*Description*
Pole	(O)	The centre of the mirror.
Principal axis	(XY)	The straight line passing through the centre of curvature and the pole of the mirror.
Focal point	(F)	The point on the principal axis at which incident rays parallel to the principal axis: (a) intersect after reflection from a concave mirror, or (b) from which parallel incident rays appear to diverge after reflection from, a convex mirror.
Focal length	(f)	The distance between the pole and the focal point.
Radius of curvature	(OC)	The radius of the sphere with centre C of which the mirror forms a part.
Focal plane	(PQ)	The plane that passes through the principal focus perpendicular to the principal axis.

Reflection from Concave and Convex Mirrors

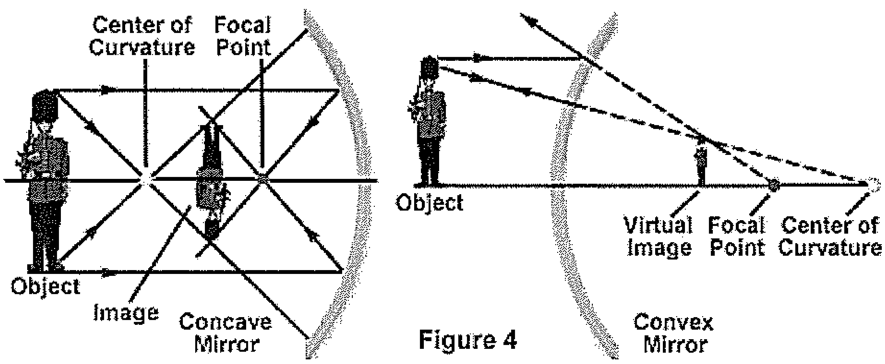


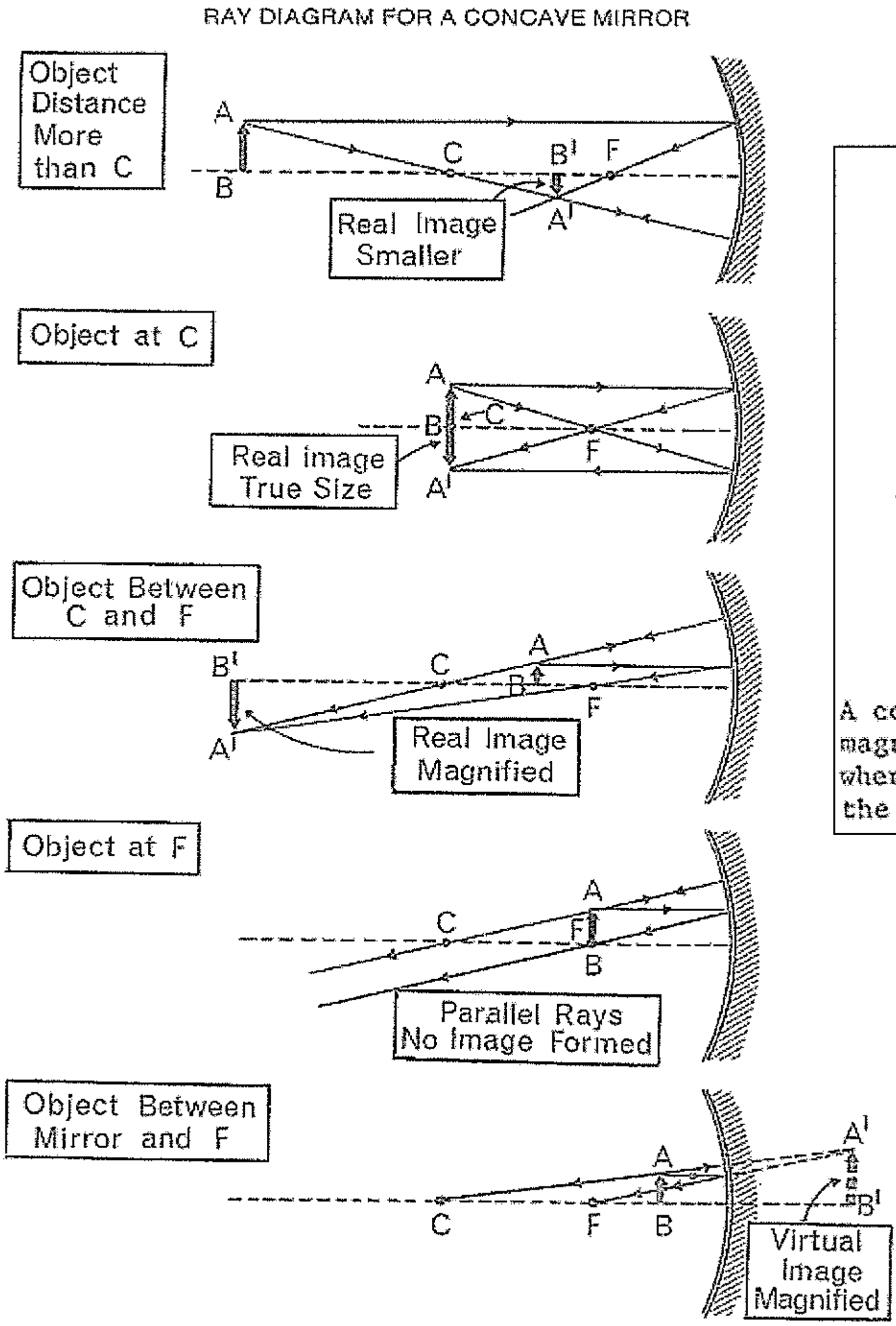
Figure 4

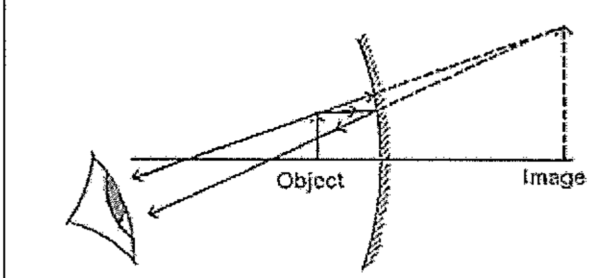
Concave mirrors

➔ Concave mirrors are **curved inwards** and reflect light inwards.

➔ They are also known as **converging mirrors** because all the rays are reflected towards a common point called the principal focus (converge at a focal point). The image is **enlarged, right side** up and **laterally reversed**. They are used in makeup mirrors.

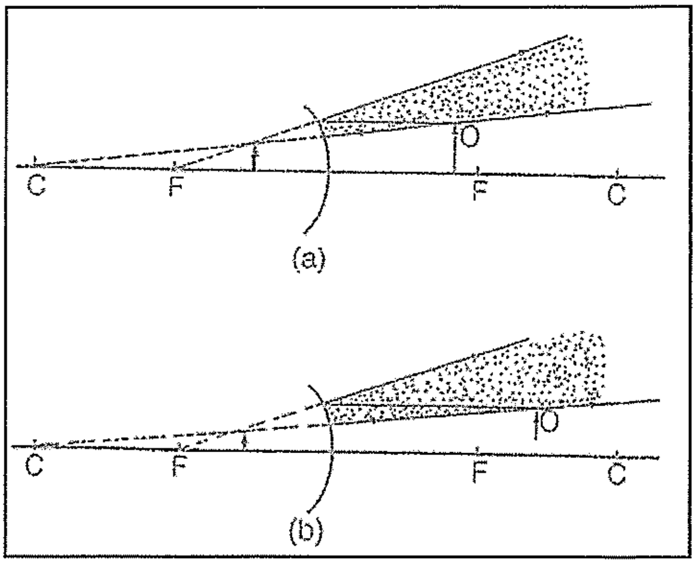
➔ Concave mirrors are used in flash lights and car head lights. Telescopes use concave mirrors to focus light from distant stars and very dim objects in the sky. Concave mirrors are also used as dentist's mirrors and shaving mirrors because of their ability to produce highly magnified up right images.





A concave mirror will produce a magnified, upright, virtual image when the object is placed between the focal point and the mirror.

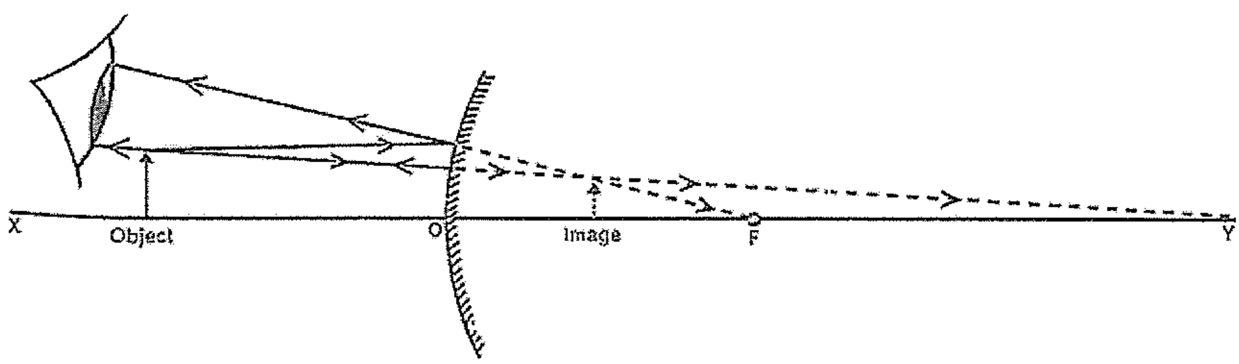
Convex mirrors



➔ Convex mirrors are **curved outwards**. They are called **diverging mirrors** because the reflected rays move away from each other (diverge) and do not meet. The image is smaller, the right side up and provides a wider view.

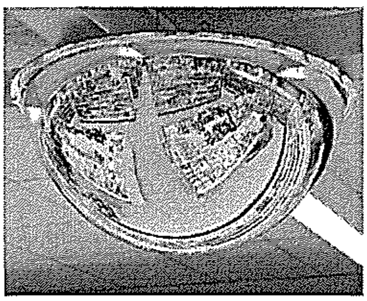
➔ Look at the diagram on the right. The image is always **virtual, diminished** (smaller) and **upright**.

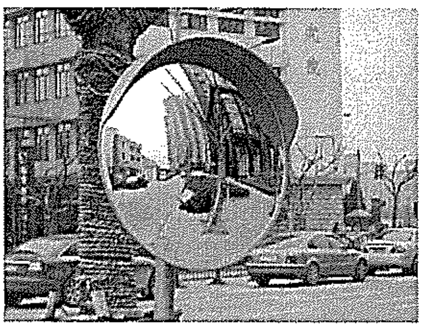
➔ It is used in rear-view mirrors of cars at corners of roads and on shop and bank ceilings. Because the image always appears smaller in convex mirrors, these mirrors are able to provide a wider view of the road behind or the shop or the interior of a bank.



A convex mirror always produces a diminished, upright virtual, image.

What would drivers have to keep in mind when using these mirrors? (Think about the warning sign written on them).





DEFINITIONS

What do these words mean? Use diagrams to explain your answers. Use 8⟶ as your symbol.

Real image - ____

Virtual image - ____

Diminished - ____

Magnified - ____

Inverted - ____

Laterally inverted - ____

Upright - ____

Demonstration – Reflections in mirrors

Materials – plane, convex and concave mirrors

Starting with the mirror at arm's length, slowly bring the mirror towards your eye. Describe any changes in the images (use the words you defined above in your observations).

Plane mirror –

Concave mirror –

Convex mirror –

Place a plane mirror above and at the ends of the words below.

CARBON DIOXIDE

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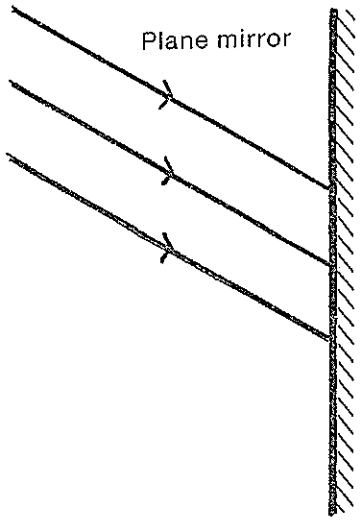
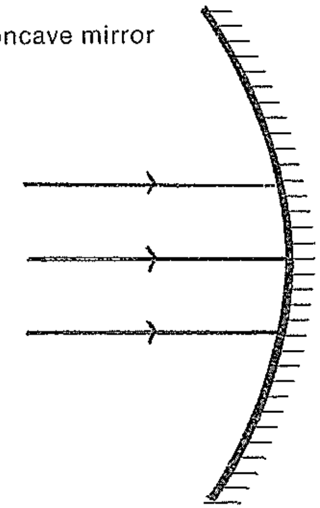
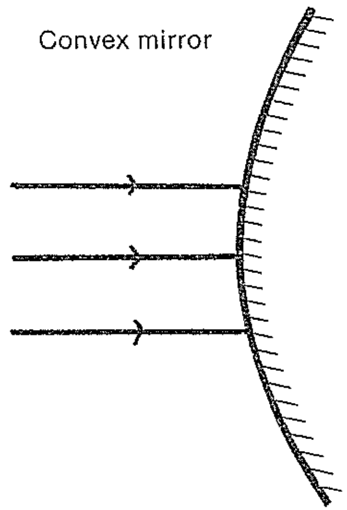
Briefly describe what you observed -

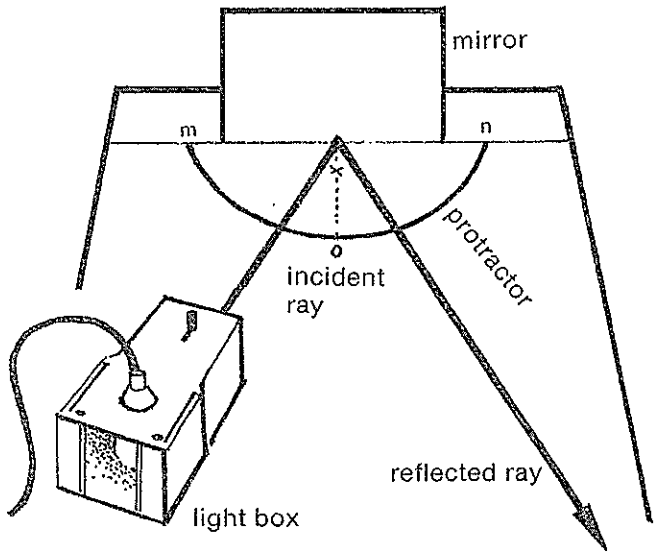
MIRRORS ANGLES REFLECTION

TRACE THE RAY









MEASURE THE ANGLE

PROCEDURE

Using the prepared protractor, place the plane mirror so that its reflecting surface is on the line MN.

Using a single ray of light, start at O and position the light box at various angle between O and M. The angle between OX and the incident ray is called the incident angle and is recorded in a table of results. The angle of reflection (the angle between OX and the reflected ray) is also recorded.

RESULTS

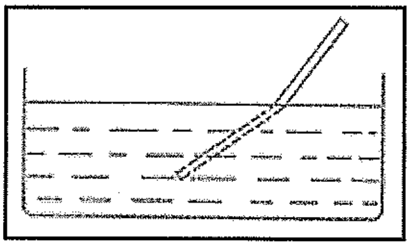
ANGLE OF INCIDENCE	ANGLE OF REFLECTION

CONCLUSION

REFRACTION

➔ **Refraction** – When rays of light pass through one transparent substance to another, they **change direction** at the boundary between the two substances. This is called refraction.

Demonstration *– place a pencil on end in a glass of water, the pencil will appear bent or even broken. This is due to refraction. The light rays are bent or refracted here because water slows down the light rays. A result of this change in speed is that rays are bent or slanted into a new path.*



➔ Similar bending takes place when light passes from **air to glass.**

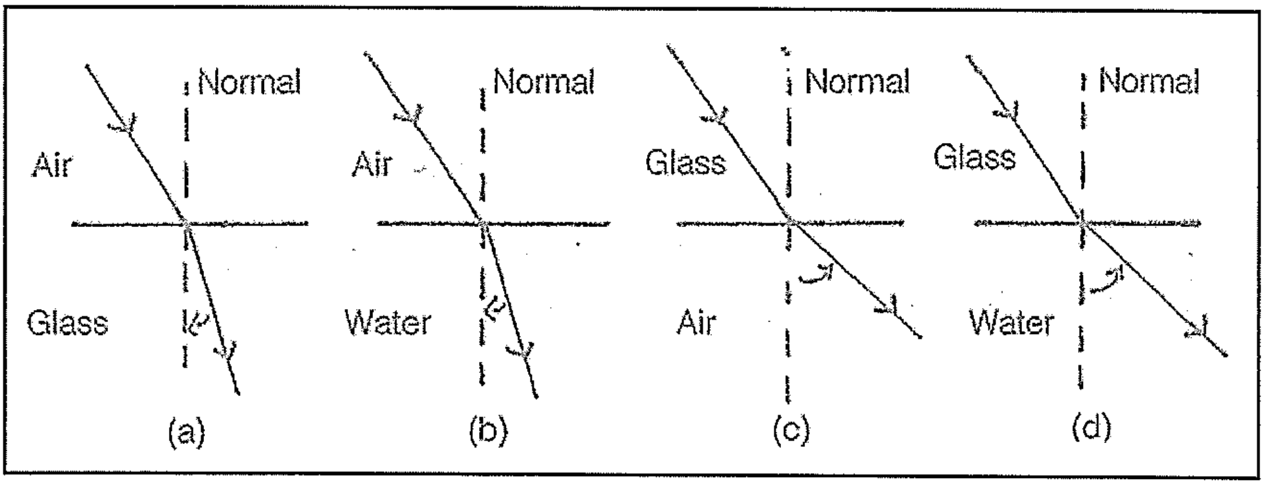
➔ When light rays travel from water to glass or glass to air they **speed up** and the bending is in the opposite direction from what it is when light rays are slowed down.

➔ Water is **optically denser** than air.

➔ Glass is **optically denser** than water.

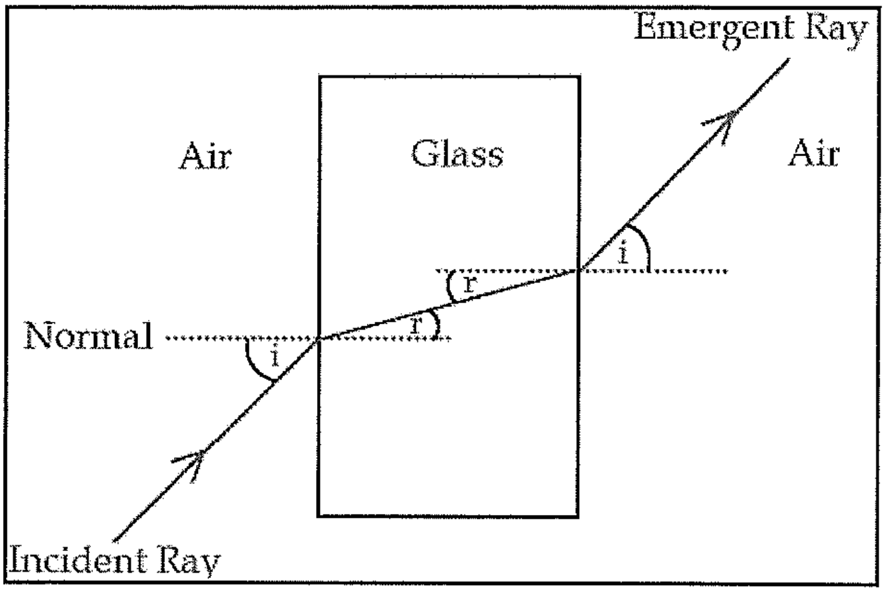
➔ Look at the diagram below. When a ray of light passes at an angle from an optically less dense medium to an optically more dense medium (e.g. air to water, water to glass), the ray is refracted **towards the normal** (a and b on the diagram).

➔ When a ray of light passes at an angle from an optically more dense medium to an optically less dense medium (e.g. glass to air, glass to water) the ray is refracted **away from the normal** (c and d on the diagram).



➔ When a ray of light passes from air into glass and then back into air, it refracts **twice**.

➔ Look at the diagram on the right. The emergent ray or the ray that comes out of the glass is parallel to the incident ray or the ray that strikes the glass.



1. With a ruler, continue the incident ray through the glass and out to the other side. This shows you how the incident ray is parallel to the emergent ray.

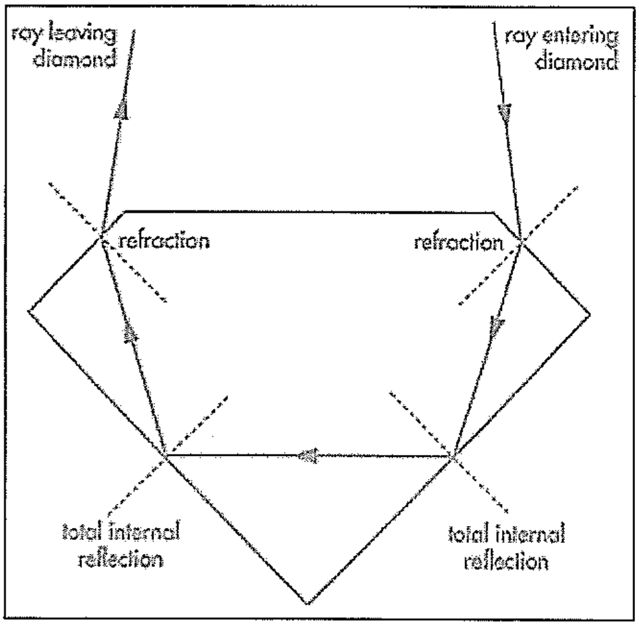
2. What does the 'i' represent? ______________________________

3. What does the 'r' represent? ______________________________

Total internal reflection

➔ Refraction can cause another tricky thing called internal reflection to happen.

➔ Sometimes as light passes from one substance to another, some of the light gets refracted **back into** the substance. When all the light gets internally refracted it is called total internal reflection. This happens in diamonds and we see it as sparkles.



Total internal refraction

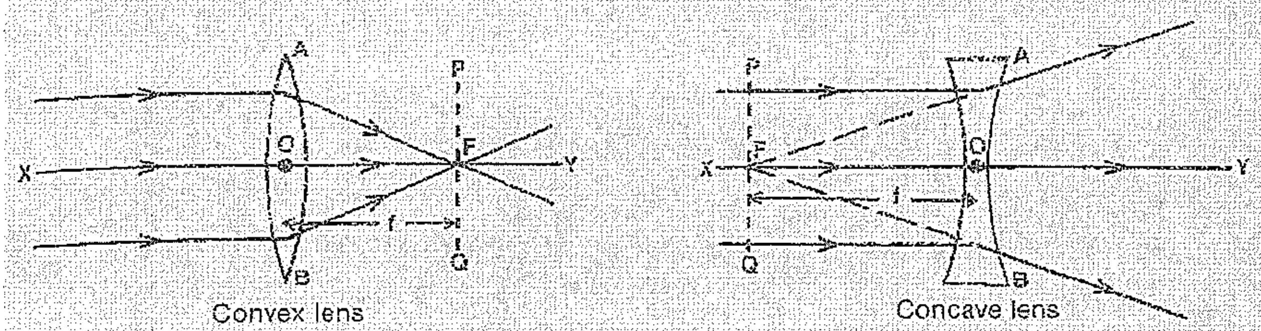
1. Look at the fish tank from the front then very slowly move to a corner of the tank.
2. Keep looking into the tank as you do this.
3. Very gently and slowly move your head from side to side while you are standing at the corner of the tank.
4. Do this a few times and then write down what you see.

____

____

LENSES

TERMS USED IN CONNECTION WITH LENSES



Term		*Description*
Optical centre	(O)	The centre of the lens.
Principal axis	(XY)	The line running through the optical centre at right angles to the plane of the lens.
Focal point	(F)	The point on the principal axis through which incident rays parallel to the principal axis: (a) converge after passing through a convex lens, or (b) from which incident rays appear to diverge after passing through a concave lens.
Focal length	(f)	The distance between the optical centre and the focal point.
Aperture	(AB)	The diameter of the lens.
Focal plane	(PQ)	The plane passing through the focal point and perpendicular to the axis.

→Lenses are **transparent** (light passes through) and have been shaped especially so that light passing through them will bend.

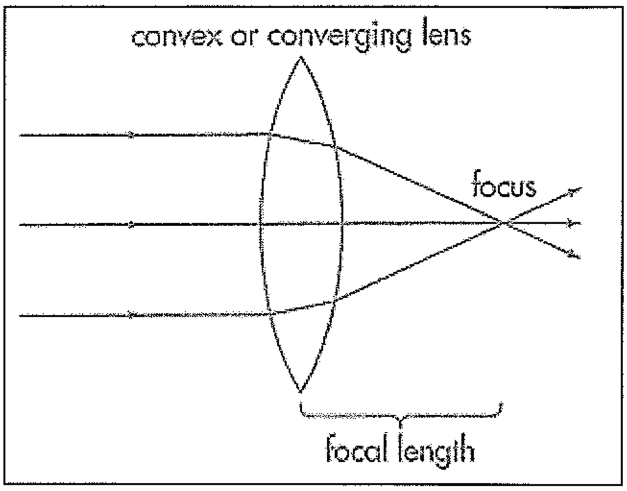
→There are two different types of lenses: **convex and concave**

Convex lenses

→ Convex lenses are called converging lenses because they make the light come together or focus at a point.

→ When light from something that is far away passes through a convex lens onto a screen an image can be seen. This image is upside down.

→ If you hold a convex lens near an object then the object is seen the right way up and is larger or magnified.



USING SCIENCE FOCUSING HEAT

As well as bringing light rays to a focus, a convex lens will concentrate the infrared (heat) rays from the Sun at a point. If you focus the light from the Sun on to a piece of dark paper you may be able to get it to smoulder. Be very careful not to burn yourself—and **never look directly toward the Sun itself. This will permanently damage your eyes!**

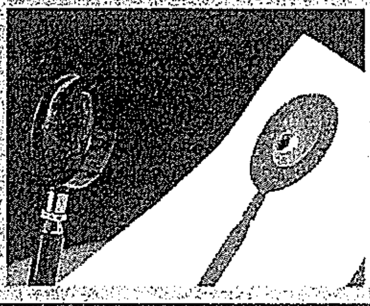


FIGURE 7.9
A lens can focus the Sun's heat rays and cause a fire to start.

Write down all the uses of convex lenses that you can think of.

___

___

Write your observations below. How close did the magnifying glass have to be to the paper to start it burning?

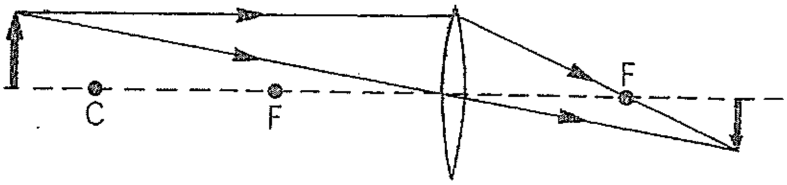
___

Try other colours of paper. Write your observations below.

___

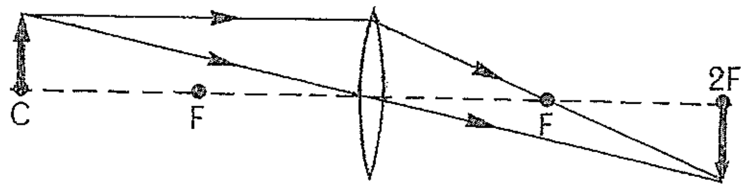
RAY DIAGRAMS FOR A CONVEX LENS

Object outside 2F



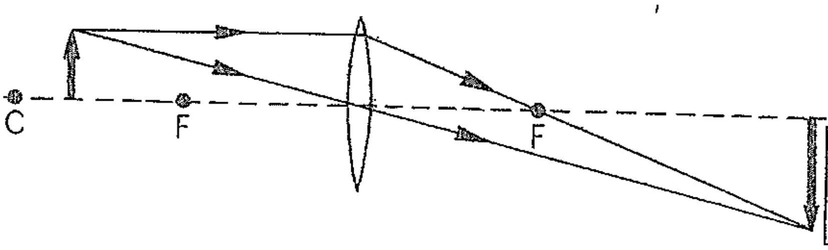
Real Image Smaller

Object at 2F



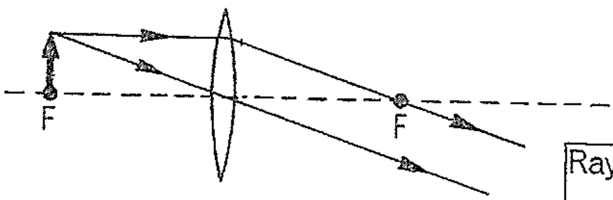
Real Image True Size

Object between F and C



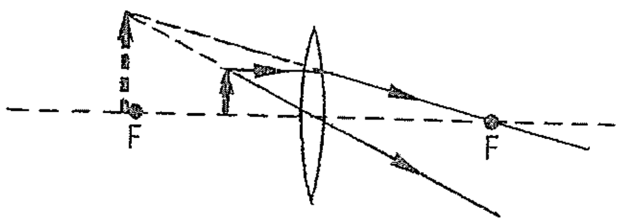
Real Image Enlarged

Object at F



Rays Are Parallel- No Image

Object between F and lens



Virtual Image Enlarged

Rays Diverge

Concave lenses

→ Concave lenses are called **diverging** lenses.

→ Diverging lenses make the light that passes through them **spread** out. Concave lenses make images that are **smaller** and the **right way up**.

→Concave lenses are also very useful. Concave lenses work by making the light bend or refract outwards as it passes through.

In the space below, come up with a way to remember the different shapes of concave and convex lenses.

Demonstration – Magnifying glass

Equipment – magnifying glass and concave lens.

1. Hold a magnifying glass up and look at objects
Up close
From a distance
*Say what type of lens this is by looking at its shape*____________________
2. Try it again but this time use a concave lens instead of the magnifying glass.
3. List any similarities or differences.

LENSES PRACTICAL

Materials – thin and thick convex lens, thin and thick concave lens, light box.

1. Place a thin convex lens below. Trace around it and then place the light box on the side. Using the three slotted slide, focus the three rays on the side of the lens and draw them on the diagram. Label what you have drawn.

Measure the distance of the focus from the centre of the lens. This is your focal length.

2. Place a thin concave lens below. Trace around it and then place the light box on the side. Using the three slotted slide, focus the three rays on the side of the lens and draw them on the diagram. Label what you have drawn.

Measure the distance of the focus from the centre of the lens. This is your focal length.

3. Place a thick convex lens below. Trace around it and then place the light box on the side. Using the three slotted slide, focus the three rays on the side of the lens and draw them on the diagram. Label what you have drawn.

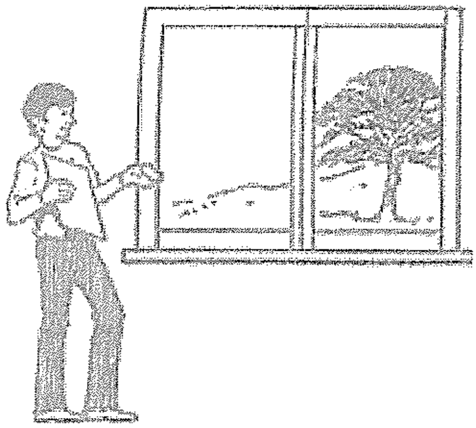
Measure the distance of the focus from the centre of the lens. This is your focal length.

4. Place a thick concave lens below. Trace around it and then place the light box on the side. Using the three slotted slide, focus the three rays on the side of the lens and draw them on the diagram. Label what you have drawn.

Measure the distance of the focus from the centre of the lens. This is your focal length.

LENSES AND IMAGES PRACTICAL

Part one



1. Stand 2 or 3 metres from a window and hold the sheet of paper and a convex lens as shown in the diagram.

2. Move the sheet backwards and forwards to see if you can produce an image on the page.

3. Carefully record your observations below.

____

____

____

4. Try the procedure again, using a concave lens.

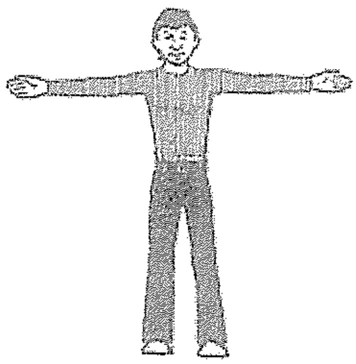
5. Record your observations.

____

____

____

Part two



1. Ask a member of your group to stand approximately 10 metres away from you. Holding the **convex** lens at arm's length, record the appearance of your partner, as seen through the lens. Then ask your partner to carry out the following.

(When you are recording your observations, use your left and right hand when describing which way the image moves.)

a) Move his or her hands upwards

____

____

b) Move the left then right hand

____

c) Now ask your partner to walk towards you, stopping 10cm away from the lens.

Carefully record your observations.

Repeat this procedure, using a **concave** lens.

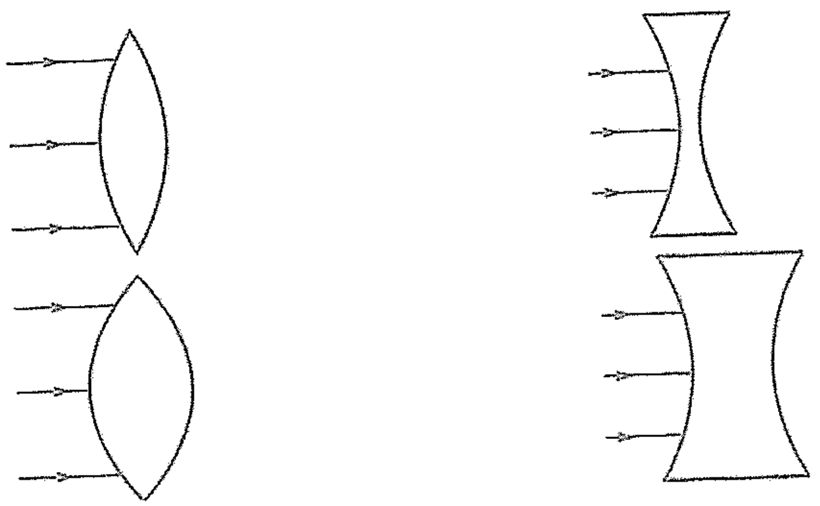
a) Move his or her hands upwards

b) Move the left then right hand

c) Now ask your partner to walk towards you, stopping 10cm away from the lens.

Carefully record your observations.

Complete the diagrams to show how the light refracts through the convex and concave lenses. Label the focus and the type of lens it is.



COLOURS OF LIGHT

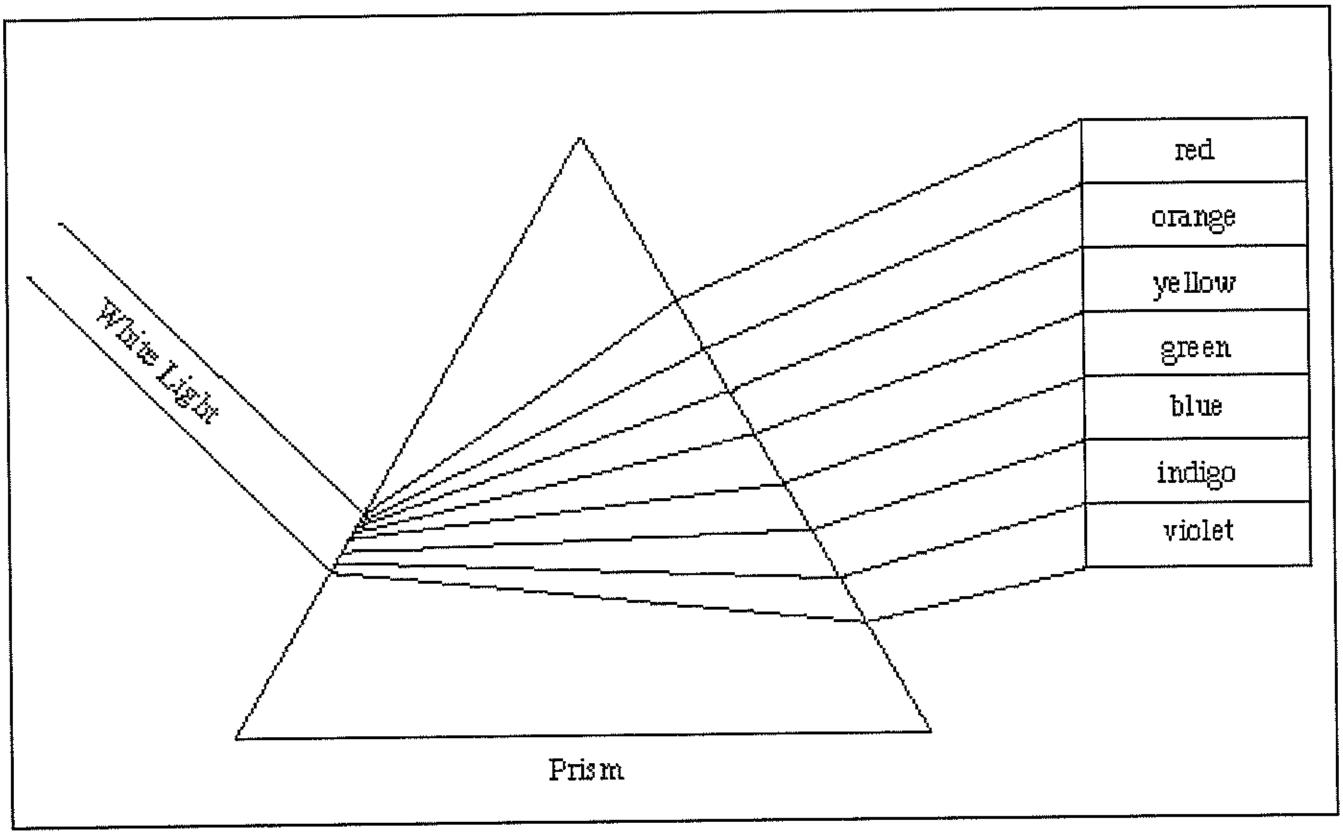
➔ In 1666, Sir Isaac Newton observed a band of colours appearing on a screen when he shone a beam of light through a prism.

➔ The colours were **violet, indigo, blue, green, yellow, orange** and **red**.

➔ From this Newton concluded that **white light** was a mixture of various coloured mixed in certain proportions.

➔ He called this band of colours a **spectrum** and the splitting of white light into colours is called dispersion.

Colour in the spectrum below in the correct colours.



Demonstration – Creating a spectrum

Materials – light box, convex lens, piece of white paper, triangular prism.

1. Arrange the light box so that a single slit of light shines upon a triangular prism.

2. On the other side of the prism hold a screen and try to create a spectrum onto the screen.

3. Record what you see and in order of colours.

____

____

____

4. Put in a red filter, what do you see now?

____

____

→ Lights can pass through objects or be reflected by objects.

→ Light can also go into objects and not come out – this is called being absorbed.

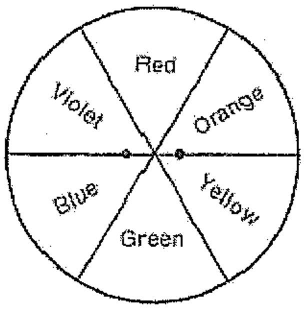
→ So when light hits something it can:

- Pass **through** the object – for example ______________________.
- Be **reflected** by the object – for example ______________________.
- Be **absorbed** by the object – for example ______________________.

1. White light is made up of **seven** different colours of light.
2. A **prism** is a special lens that separates white light.
3. If an object reflects yellow light, you see the object as yellow.
4. The colours that an object absorbs are the colours that it takes in.
5. A green leaf absorbs all of the colours except for green.
6. You seen an object as the colour of the light it **reflects**.

In the space provided draw a picture of a red apple. Show all the colours of white light hitting the apple like a rainbow. Show that the only colour being reflected is green.

Demonstration – Newton's disc.



Materials – cardboard, colouring textas, scissors, ruler, pencil, white paper

1. *Use the bowl to trace a circle onto the cardboard. Cut it out.*
2. *Cut a piece of paper the same size and glue it to the cardboard.*
3. *Divide the white disc into six segments and colour the segments .*
4. *Carefully make two small holes (1/2cm) each side of the centre of the disc. Thread 1 metre of string through the holes, knotting it to form a loop.*
5. *Holding the disc as in the diagram, twist the string by moving one hand. When this has been done, move your hands apart and then inward as the string winds up. With a little practice you should be able to make the disc appear a whitish colour.*
6. *What do you observe and explain why it is happening.* ______________________

____

____

____

____

____

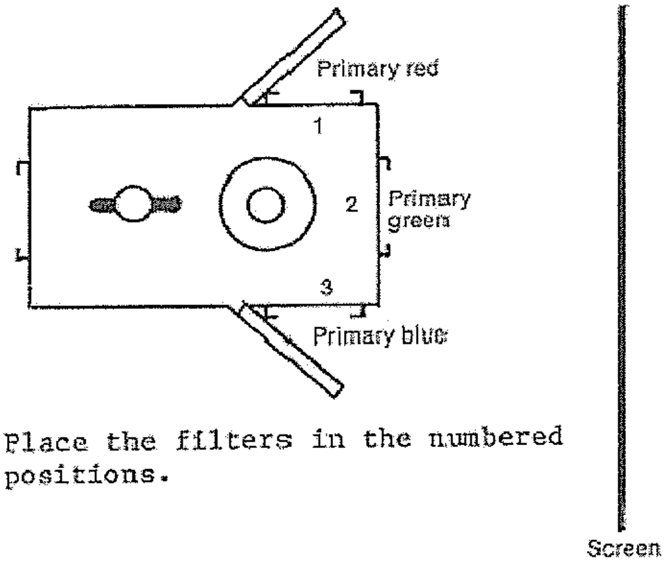
____

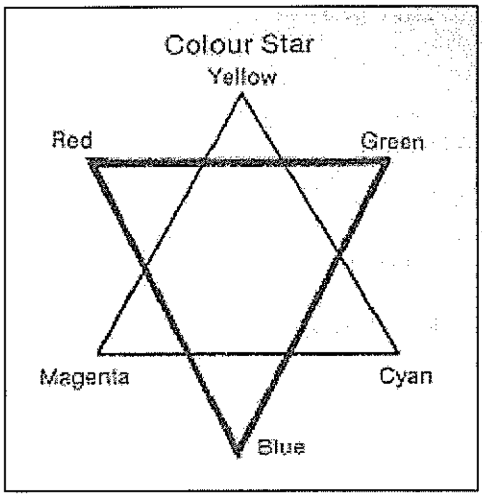
____

MIXING LIGHT PRACTICAL

Materials – light box, white screen.

Procedure – By closing the side mirrors and blocking out the end filter of the Hodson light box combine the colours in as many ways as possible.





Colours mixed	Description of colours on screen

1. Look at the colour star. Compare your observations to this star.

____

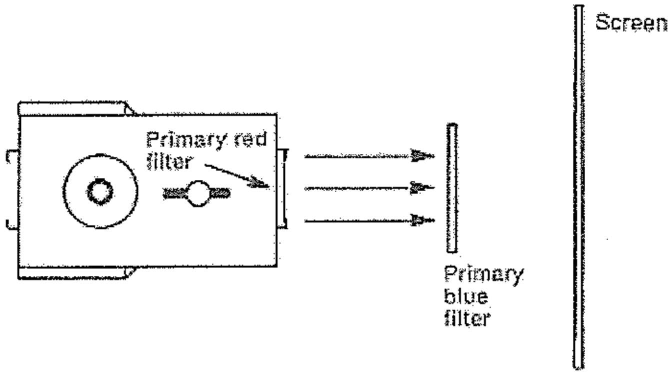
2. What is meant be the term 'primary colours'

____

____

COLOURED FILTER PRACTICAL

Set up the Hodson light box with the red and blue filters, as shown in the diagram. Record what you see on the screen.



Change the colour of the light by changing the filter in the end of the light box. Record your observation in the table.

Coloured filters – transparent object

Colour of light	Colour of filter	Colour on the screen
Primary red	Primary blue	
Primary red	Primary yellow	
Primary blue	Primary green	
Primary green	Yellow	

Coloured filters – opaque object

Using a filter in the end of the light box, shine coloured light onto the pieces of coloured cardboard.

		COLOUR OF OBJECT			
		White	**Red**	**Green**	**Blue**
COLOUR OF LIGHT	***White***				
	Red				
	Green				
	Blue				

Match the following terms to their definitions.

Translucent, opaque, concave, transparent, lens, law of reflection, convex, mirror, Electromagnetic spectrum, prism, refraction.

________________________ - Allowing light to pass through with almost no distortion.

________________________ - Transparent material with at least one curved surface.

________________________ - Some light can pass through and some is blocked.

________________________ - A range of all light waves.

________________________ - Does not allow any light to pass through.

________________________ - Light rays strike and leave a mirror at equal angles.

________________________ - A thick piece of glass that bends light.

________________________ - A polished surface that forms reflective images.

________________________ - A surface that curves inward.

________________________ - The bending of light as it passes between two transparent materials.

________________________ - A surface that curves outward.

1. A material or object that blocks light completely is ________________.

2. How does light change when it enters a new medium?

____

3. Which kind of light has a wavelength shorter than green light?
   a. Red light
   b. Radio waves
   c. X-rays
   d. Yellow light

4. Explain how light travels: ____

____

5. You would choose a ______________ material if you wanted light to pass completely through it.

6. How is refraction similar to reflection?

____

____

____

____

7. Which light has the most energy?
   a. Radio waves
   b. X-rays
   c. Gamma waves
   d. Microwaves

8. A blue shirt looks blue because
____.

9. A yellow dress looks yellow because
____.

10. The range of different forms of electromagnetic radiation, arrange by wavelength, is called the ____________________________

11. What does an infrared sensor detect?
____

12. The primary colours of light are red, blue, and __________.
   a. Cyan
   b. Yellow
   c. Magenta
   d. Green

13. Explain what determines an object's colour.

____

THE EYE

➔ The eyes are perhaps the most delicate organs in your body.

➔ The eye is a **living camera**. It has a lens which prints its pictures in our brains and in our memory.

➔ It teaches us more about our environment than all our other senses put together.

Structure of the eye

➔ The eyeball is the main part of the eye.

➔ Just as a convex lens forms an image which can be focused onto a screen, the lens of the eye can focus an image onto the retina.

➔ When we say a person has blue eyes or brown eyes we are talking about the iris. The iris controls the amount of light entering the lens of the eye.

➔ The white of the eye is called the sclerotic coat which covers almost 85% of the eyeball. It is opaque and tough.

➔ The cornea is the remaining part of the eye and is transparent to allow light to reach the lens and the interior of the eyeball.

➔ The space behind the lens is filled with a jelly-like substance called the vitreous humour.

➔ A liquid called aqueous humour fills the space in front of the lens.

Demonstration – Eye pupils

In partners, look at each other's eyes. Both close your eyes shut tight for a minute and then open. Look at each other's eyes and record your observations.

____

____

____

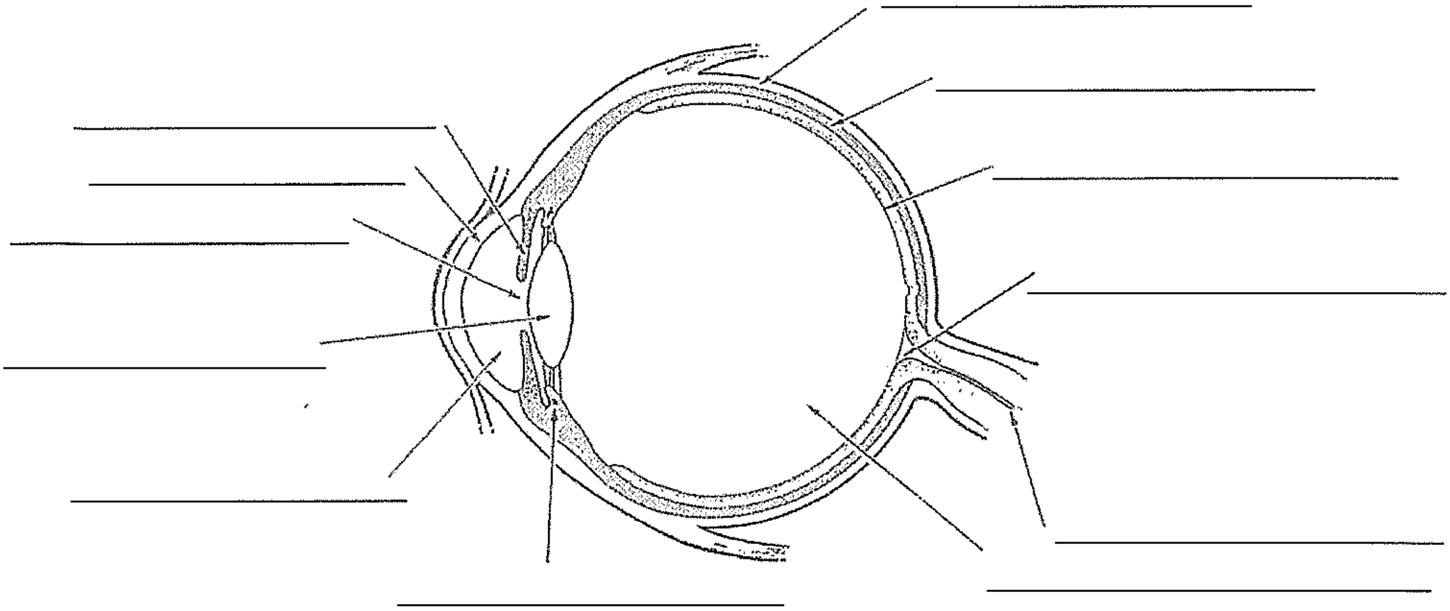
➔ The choroid coat bulges in front of the eye and includes the ciliary muscles which hold the lens.

➔ In the centre of the lens, is the dark opening of the eye called the pupil. Through this pass the light rays. The pupil's size is controlled by the ciliary muscles.

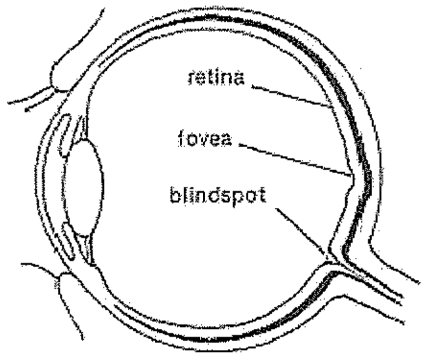
- A **bright light** cause the pupil to **contract**.
- A **dim light** causes the pupil to **open** as wide as possible.

→ At the back of the eye is the **retina** upon which the lens **focuses images**. The retina has many nerve endings which receive all the rays coming through the lens and sends these impressions to the brain via the optic nerve.

- The image brought to the brain is **inverted**. In the brain, the image is interpreted in an **upright position**.



→ The **retina** (soft, transparent skin that covers the inner back of the eyeball) is made up of several layers.



→ One of these layers is made up of rods and cones.

→**Cones** – there are about 6 million cones in the human retina

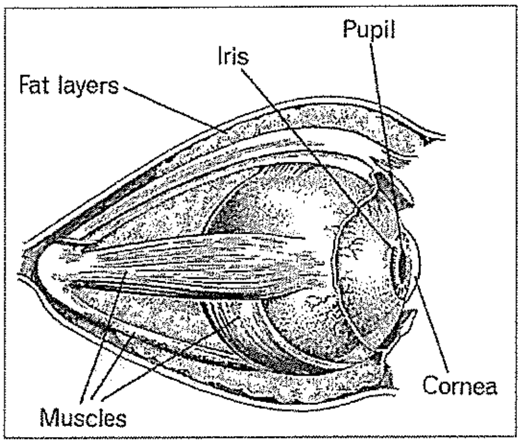
- Cones are cone shaped.
- The cones detect the fine lines, composition of an image and colour.
- The cones in your eye make it possible for you to read these words.

→ **Rods** – there are about 115 million rods in the human retina.

- Rods are rod shaped and can only sense light and dark tones of an image.
- The sensitive rods can distinguish outlines or silhouettes of objects in darkness.

→ The **blind spot** - where the optic nerve joins the retina, there are no light-sensitive cells.

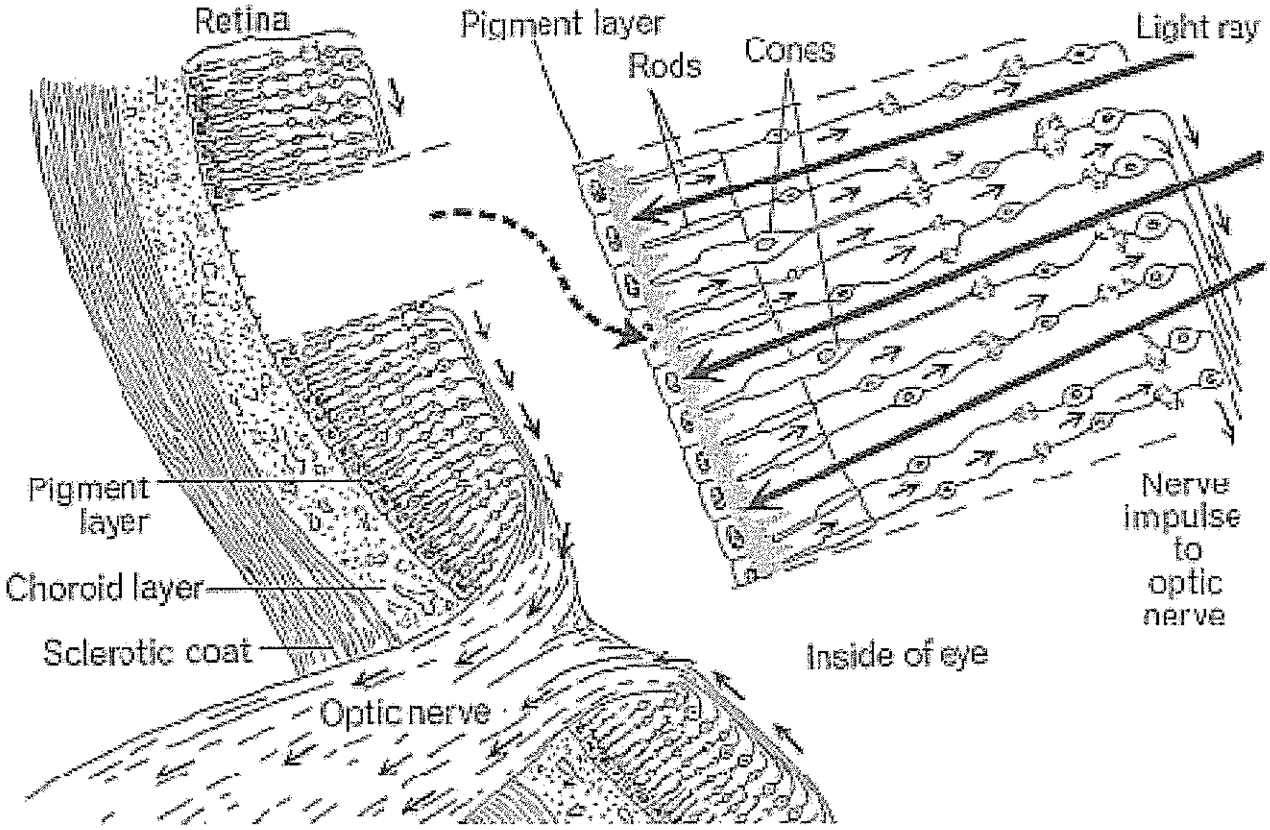
➔ Thus, any image focused on the blind spot will not be seen. Fortunately this is no disadvantage to most of us, for what is not seen by the left eye will be seen by the right and vice versa.



➔Images detected by the cones or rods are transmitted to the vision centre of the brain via the **optic nerve**.

➔The **tiny inverted** images focused by the lens onto the retina are interpreted by the brain and we see them as **upright and life size**.

➔We have to **learn** to see and we spend the first few years of babyhood and childhood learning to do this.



Normal vision

➔ When the eyes look at an object, the image formed on the retina should be sharp and clear.

➔ The eye is able to see very distant objects like the stars in the night sky as well as the printed words in a book almost at once.

➔ **Accommodation** - In order for the images to be formed sharply and clearly, the lens has the power to adjust itself to focus different objects at different distances on the retina. The eye is able to change the **thickness** of its lens.

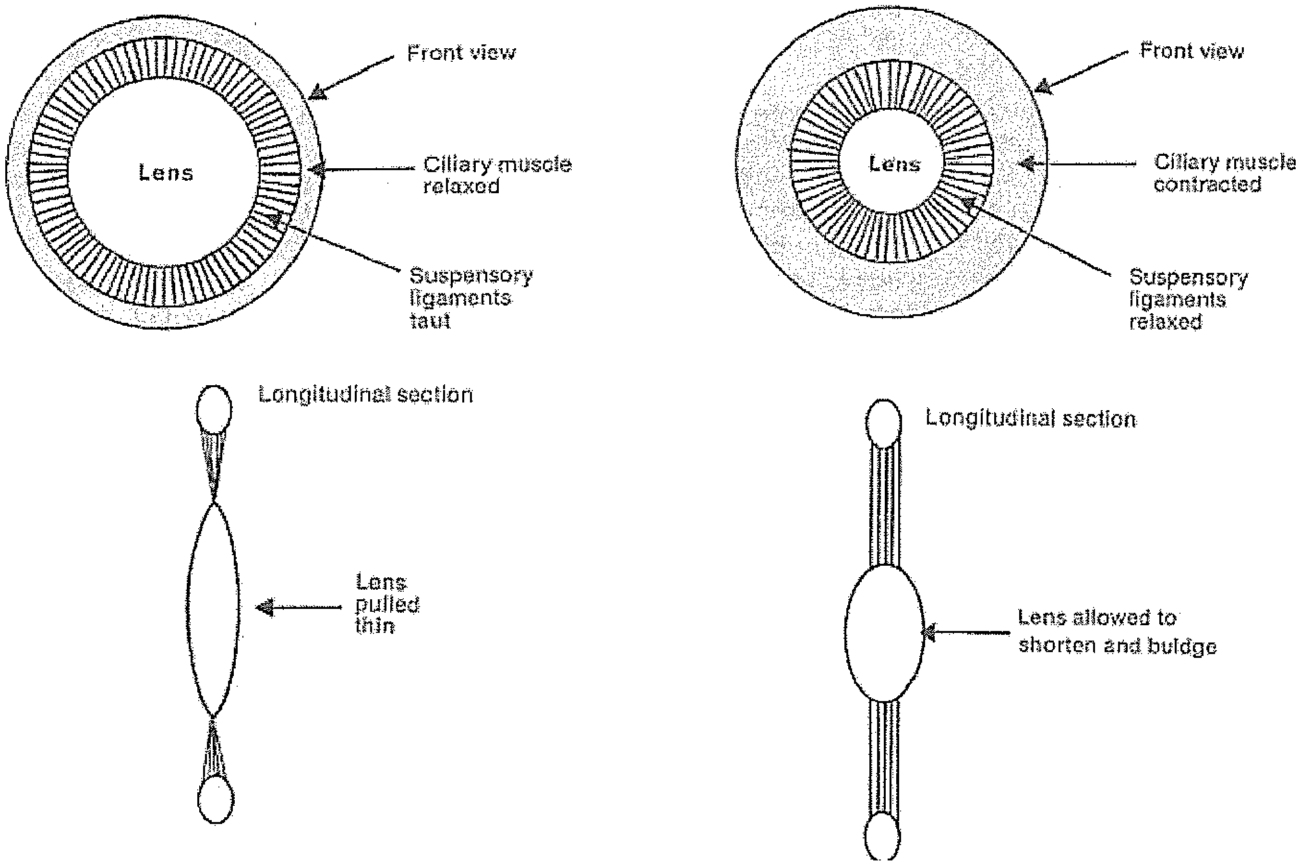
➔ When we view distant objects, ciliary muscles stretch the lens and make it **thin** = increased focal length.

➔ When we view nearer objects, the ciliary muscles press the lens and make it **thicke**r in the middle = decreased focal length.

➔ Because we have two eyes, a **double set** of visual sensations almost exactly alike is sent from objects to the signal centers of the brain, one along each optic nerve.

➔ The actual seeing takes place in the **brain**.

The diagram shows how the relaxing or contracting of the ciliary muscles increases or decreases the tension in ligaments attached to the lens. These corresponding changes in tension change the shape of the lens.

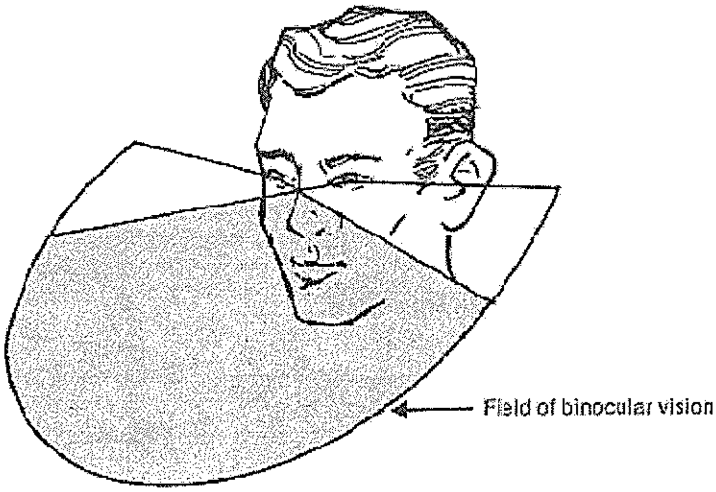


Binocular vision

➔ Normally both eyes are used when looking at an object. Therefore two images are formed, one on the retina of each eye.

➔ The brain interprets them as two view of the same object.

➔ The brain learns to judge the distance of any object by the degree of difference between the images it receives from the two eyes.



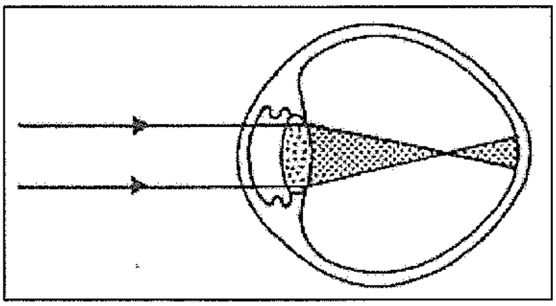
➔ This is how the brain perceives **perspective**. It estimates differences in distance between two objects or two parts of the same object. This causes a three-dimensional view.

DISORDERS OF THE EYE

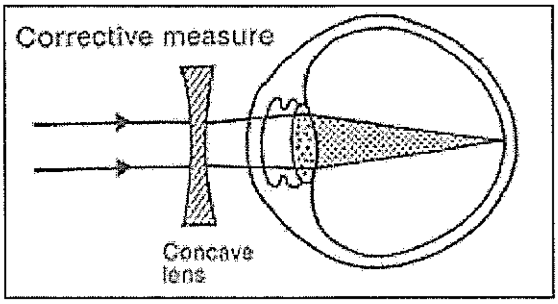
Myopia (short sightedness)

➔ In this condition the image of a distant object focuses **in front** of the retina.

➔ Myopia results from an abnormally **long distance** between the cornea and/or lens and the retina.

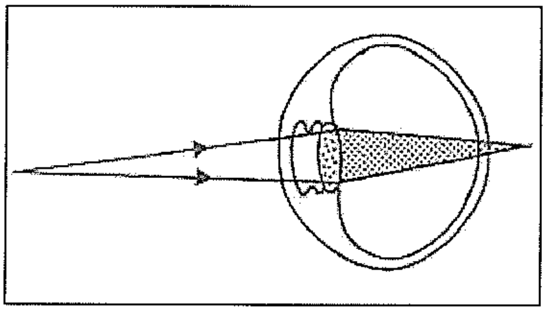


➔ This defect can be corrected by a **concave** (diverging) lens.



Hyperopia (long sightedness) or hypermetropia

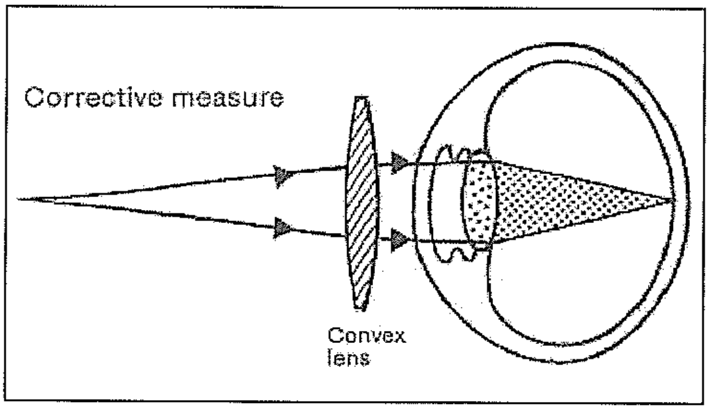
➔ In this condition the image of a close object focuses **behind** the retina.



➔ A person who is hypermetropic may initially have good vision for distance, but, depending upon the degree of hypermetropia, will eventually require glasses for distance vision as well as reading.

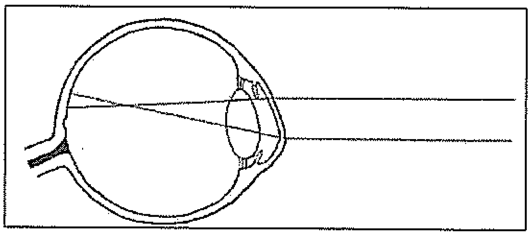
➔The reason for this is that rays of light, even coming from infinity, are unable to focus properly on the retina.

➔ This defect can be corrected by a **convex** (converging) lens.



Physical disorders of the eye

Astigmatism



➔ This condition is due to a distortion in the curvature of the lens or cornea resulting in an uneven focusing of the image.

➔ For most people, the refractive error of astigmatism becomes stabilized around the age of twenty-two, when the development of the eye is complete.

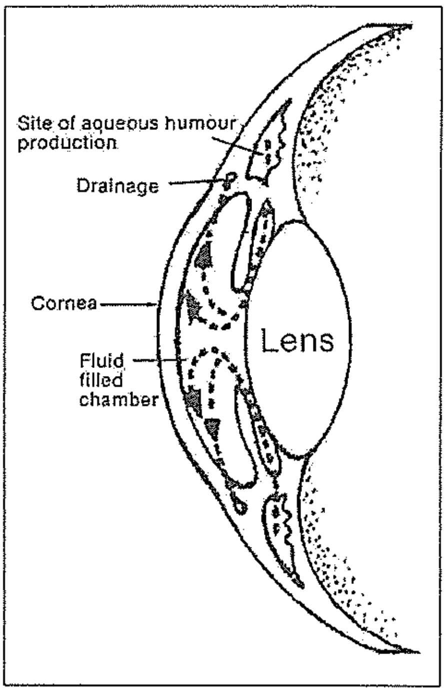
➔ This condition can be corrected by a specially designed lens that compensates for the uneven refracting ability of the eye.

Cataracts

➔In this condition the lens of the eye becomes **opaque**. The condition is common amongst older people and is the most common cause of blindness in older people.

➔ The usual remedy is to surgically remove the affected lens and replace it with a glass lens. A person then would have to wear bifocal glasses to compensate for the lack of accommodation.

Glaucoma



➔ The area between the lens and the cornea is filled with **aqueous humour**.

➔ Aqueous humour is produced in the chamber behind the iris and circulates through the pupil into the front chamber. It then drains out through pores on the edge of the cornea into a canal, eventually reaching the bloodstream.

➔ The pressure in the fluid is maintained by balancing the rate of production of aqueous humour and the rate of drainage.

➔If there is an imbalance and the rate of drainage is reduced, pressure builds up in the chambers.

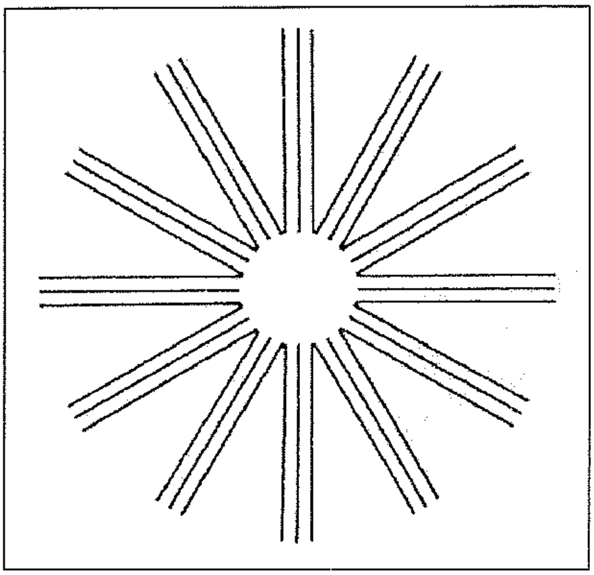
➔ Eventually this increased pressure damages the optic nerve and destroys vision.

Charts used for testing vision

➔ This chart is used to **test vision**. If people can read the second bottom line using one eye at a distance of two metres, then they have very good vision.

➔ The other eye can be tested by reading the same line backwards.

→ This chart is used to detect **astigmatism.**



→ If some of the lines appear faint to the person looking at the chart, then the person probably has astigmatism.

Colour blindness

→A set of **Ishirara** cards can be used to test for colour blindness. These cards have numbers composed of coloured dots printed on a background of coloured dots .

→ If people have defective colour vision , the numbers they see are different from those seen by a normal person.

→ The term colour blindness covers a range of inabilities to distinguish or match colours.

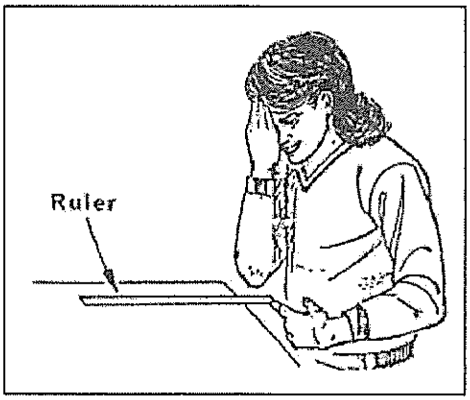
→ The condition in which people can only see objects as if they were viewing a black and white television picture is very **rare**.

→ The inability to distinguish between colours may be due to an absence of, or deficiency in, the cones; or some neurological problem causing confusion of colour perception.

→ The most common colour defect is the red -green defect ; the blue-yellow defect is much less frequent .

INVESTIGATING THE EYE

Binocular vision



1. Place a ruler so that it hangs approximately 8 cm over the end of the bench.
2. Standing directly over the ruler, place your hand over one eye and bring the index finger of your other hand from the side and try to touch the ruler. Repeat this several times.
3. Move your head a little closer and try it again.

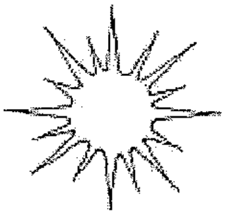
Record your observations: ____

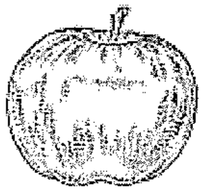
____

Blind spot

1. Hold the page vertically at arms length in front of your eyes.

2. Close your right eye and look at the apple with your left eye. Move the page slowly toward you. What happened?





Record your observations: ____

____

____

Repeat the procedure, this time close your right eye and look at the star. Was the result the same? Explain why this happens.

____

____

Explain what advantages binocular vision gives people.

____

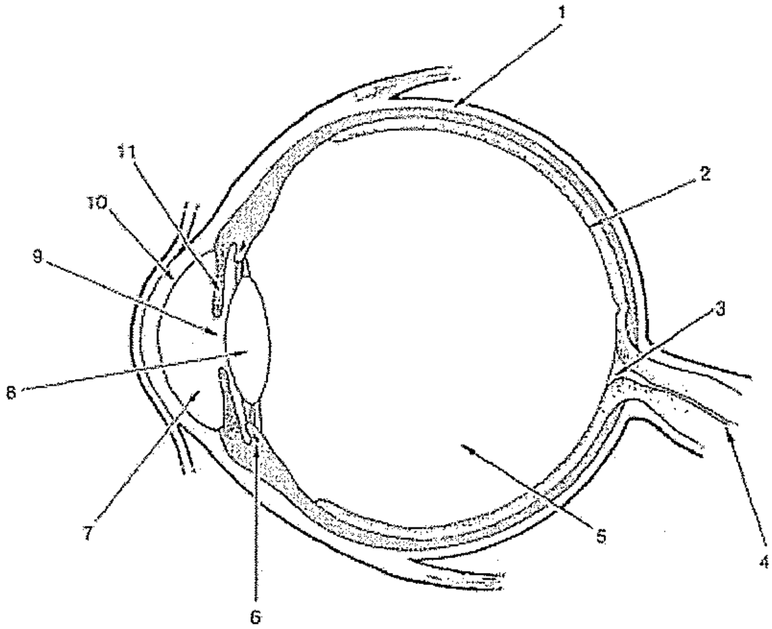
____

Briefly describe some of the problems a person with one eye would experience.

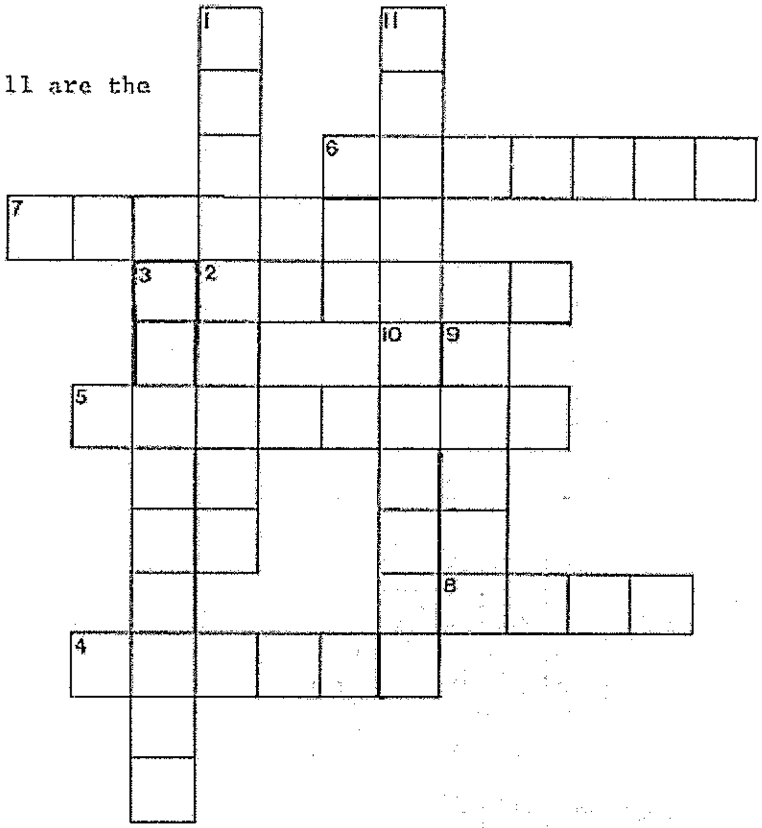
____

____

INSIDE THE EYE



The parts of the eye labelled 1 to 11 are the words in the crossword.



ACROSS

2. An image is formed here.
4. The nerve that carries the message to the brain.
5. A clear, jelly-like substance.
6. A muscle that changes the shape of the lens.
7. A watery fluid.
8. Found in both a camera and the eye.

DOWN

1. Tough coating.
3. No image is seen here.
9. It changes size, depending upon how much light there is.
10. Clear layer at the front of the eye.
11. Different people have different colours.

Fill in the table (use your own words, do not copy information directly)

Part	Structure (where is it, what does it look like)	Function (what does it do)
Aqueous humour		
Blind spot		
Choroid coat		
Ciliary muscle		
Cornea		
Iris		
Lens		
Optic nerve		
Pupil		
Retina		
Sclerotic coat		
Vitreous humour		